

Generalized Extreme Value (GEV) Analysis Equations

Stationary Model GEV Distribution Function:

$$G(z) = \exp\left(-\left(1 + \xi\left(\frac{z - \mu}{\sigma}\right)\right)^{-1/\xi}\right)$$

Stationary Return Level ($p = 1/T$):

$$z_p = \mu - \frac{\sigma}{\xi} \left(1 - (-\log(1 - p))^{-\xi}\right)$$

Non-Stationary Linear Parameters:

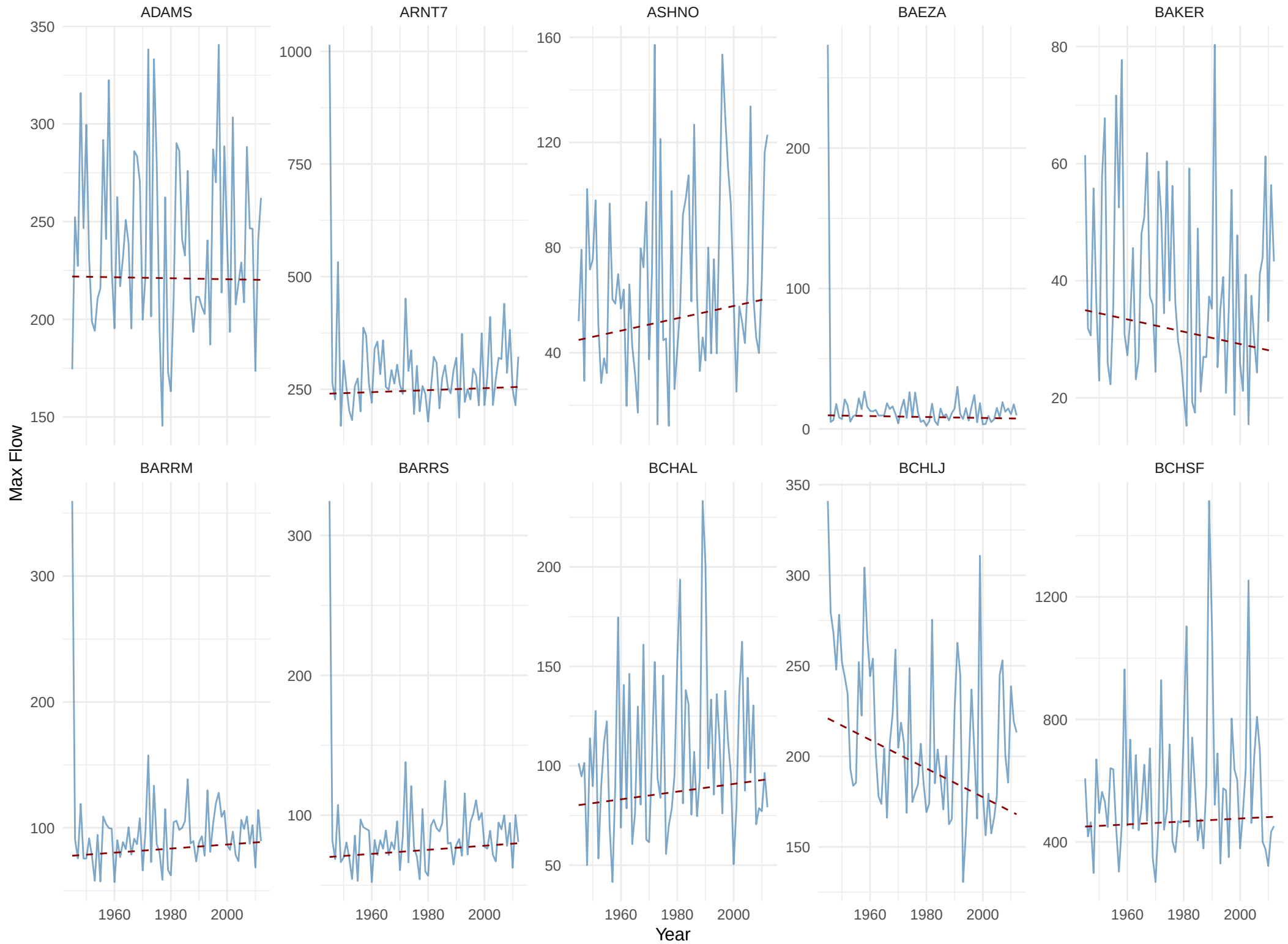
$$\mu(t) = \mu_0 + \mu_1(t - \bar{t}) \quad \sigma(t) = \sigma_0 + \sigma_1(t - \bar{t})$$

Stationary GEV Parameters and 95% CI												
Station	Stat_Mu_Est	Stat_Mu_SE	Stat_Mu_Lower	Stat_Mu_Upper	Stat_Sigma_Est	Stat_Sigma_SE	Stat_Sig_Lower	Stat_Sig_Upper	Stat_Xi_Est	Stat_Xi_SE	Stat_Xi_Lower	Stat_Xi_Upper
DAMS	221.01	5.53	210.17	231.85	40.15	3.97	32.37	47.93	-0.158	0.098	-0.350	0.034
RNT7	247.81	7.62	232.87	262.75	56.20	5.91	44.62	67.78	0.173	0.086	0.004	0.342
CHNO	52.05	3.91	44.39	59.71	27.85	2.89	22.19	33.51	-0.030	0.109	-0.244	0.184
AEZA	8.51	0.49	7.55	9.47	5.09	0.42	4.27	5.91	0.339	0.070	0.202	0.476
AKER	31.40	1.80	27.87	34.93	12.44	1.38	9.74	15.14	0.020	0.128	-0.231	0.271
RRM	83.00	2.51	78.08	87.92	18.88	1.87	15.21	22.55	0.122	0.069	-0.013	0.257
RRS	74.89	2.17	70.64	79.14	16.26	1.62	13.08	19.44	0.131	0.070	-0.006	0.268
CHAL	87.07	4.10	79.03	95.11	29.59	3.04	23.63	35.55	0.022	0.100	-0.174	0.218
CHLJ	193.30	4.84	183.81	202.79	34.92	3.54	27.98	41.86	-0.039	0.101	-0.237	0.159
CHSF	464.45	19.10	427.01	501.89	138.85	14.96	109.53	168.17	0.174	0.099	-0.020	0.368

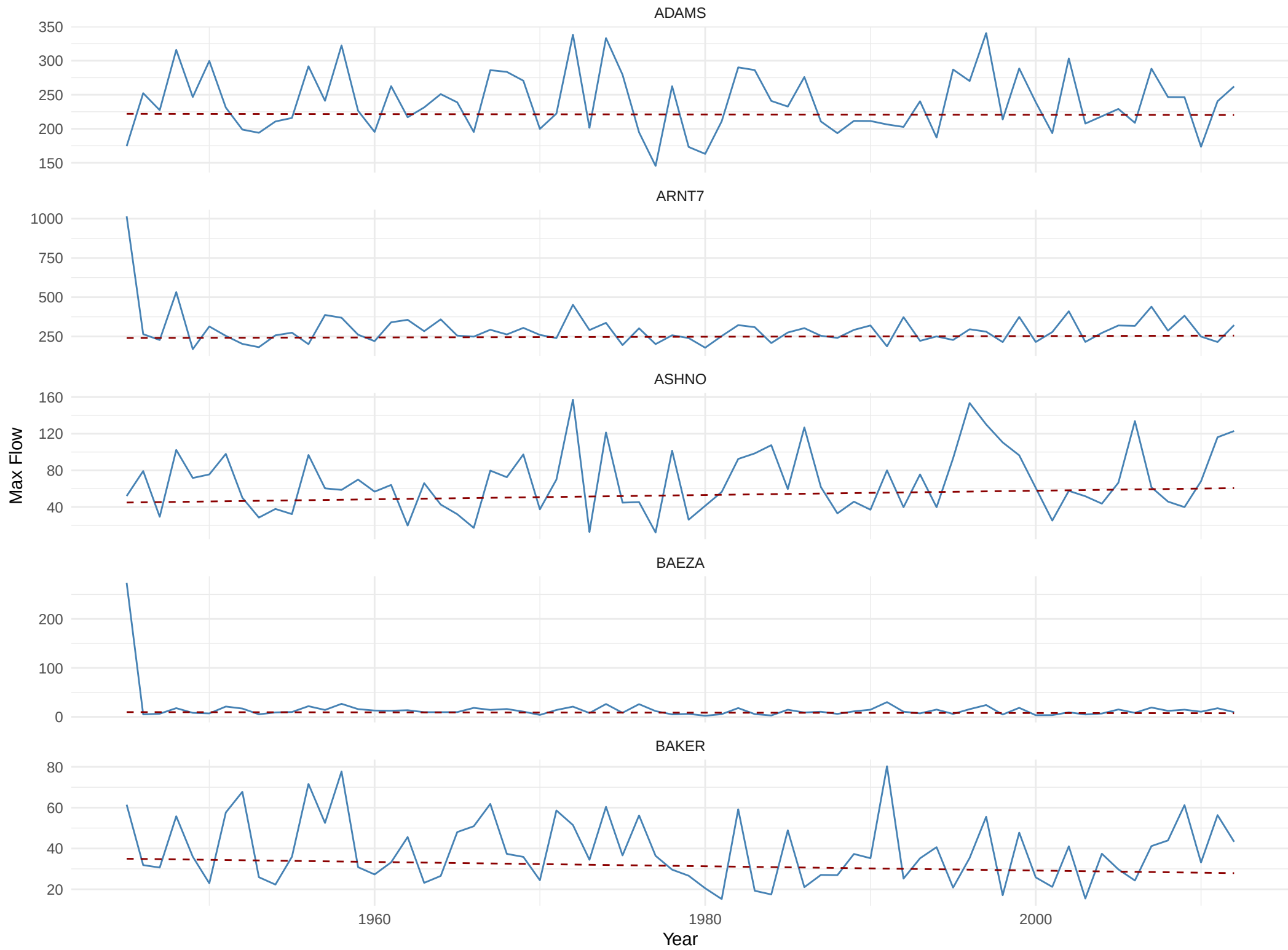
Non-Stationary Parameters and Return Levels

Station	NS_Mu0_Est	NS_Mu1_Trend	NS_Sig0_Est	NS_Sig1_Trend	NS_Xi_Est	Stat_RL_50	Stat_RL_100	NS_RL_50_Final	NS_RL_100_Final
ADAMS	221.04	−0.03	NA	NA	−0.159	337.93	352.26	329.81	343.20
ARNT7	247.82	0.22	NA	NA	0.179	560.74	642.52	505.92	572.25
ASHNO	52.73	0.23	NA	NA	−0.064	154.56	171.67	172.26	189.47
BAEZA	8.54	−0.03	NA	NA	0.336	49.83	64.86	43.88	57.09
BAKER	31.45	−0.11	NA	NA	0.024	81.85	91.30	75.55	84.56
BARRM	83.24	0.16	NA	NA	0.122	177.34	199.48	160.31	177.12
BARRS	75.08	0.15	NA	NA	0.131	157.73	177.56	144.16	159.51
BCHAL	86.71	0.19	NA	NA	0.056	207.76	230.50	210.55	234.39
BCHLJ	194.47	−0.79	NA	NA	−0.025	319.75	340.42	285.48	305.34
BCHSF	466.14	0.48	NA	NA	0.156	1239.37	1442.34	1314.11	1524.05

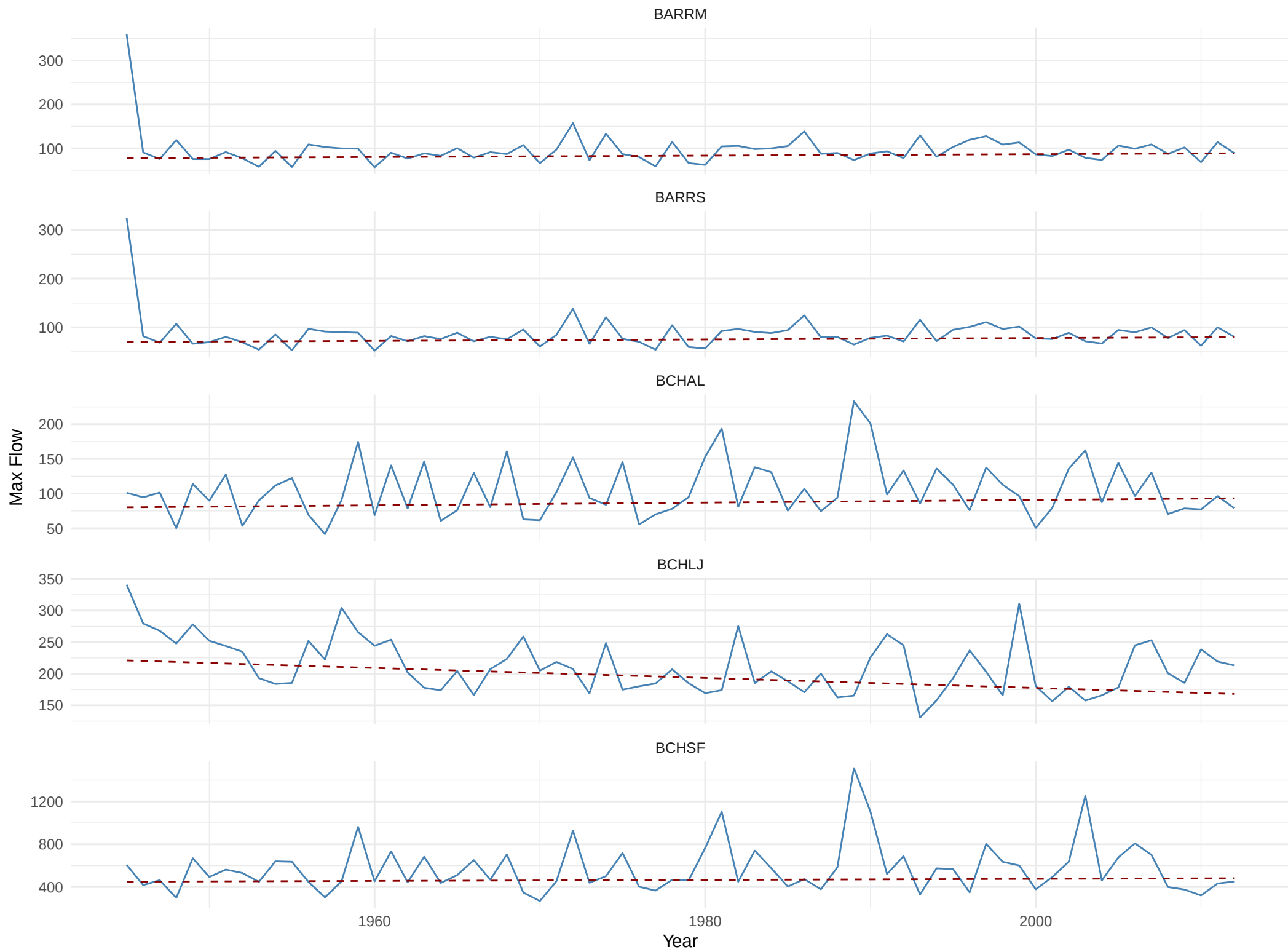
All 10 Stations: Annual Maxima vs Non-Stationary Trend (μ)

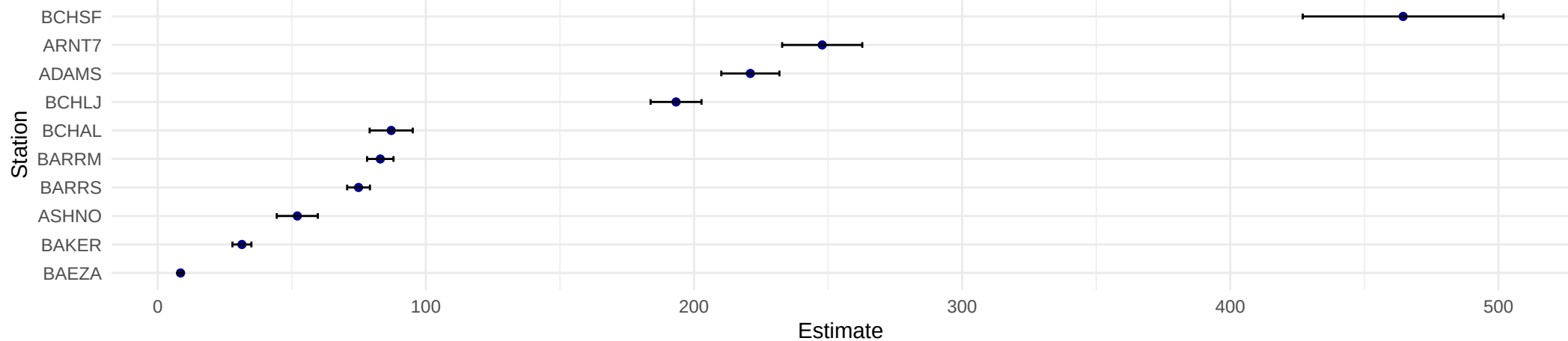
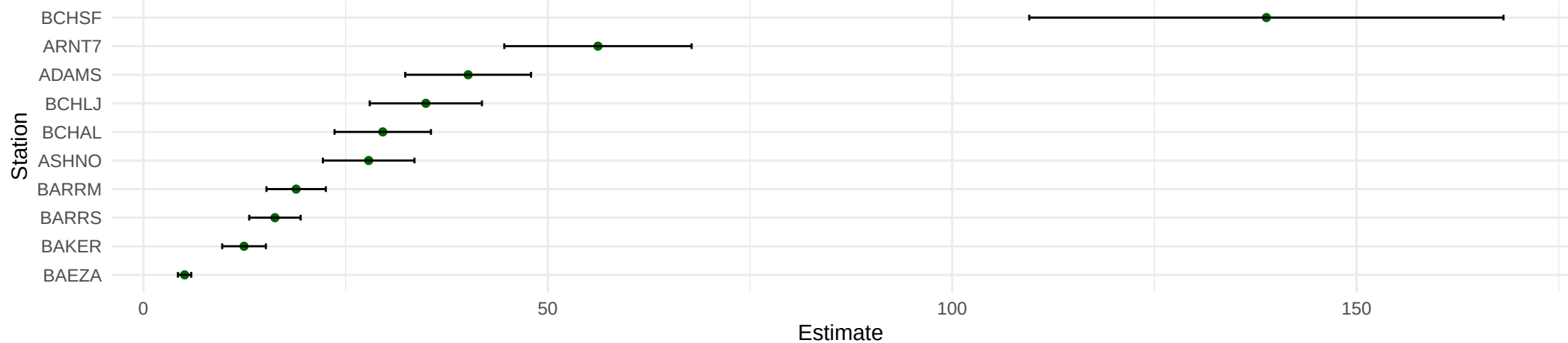
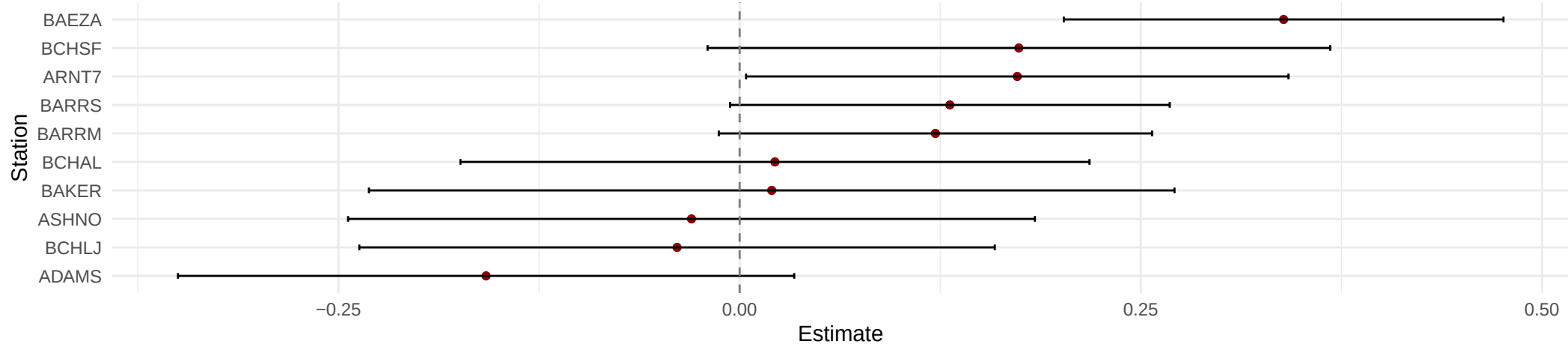


Stations 1–5: Annual Maxima vs Non-Stationary Trend (μ)



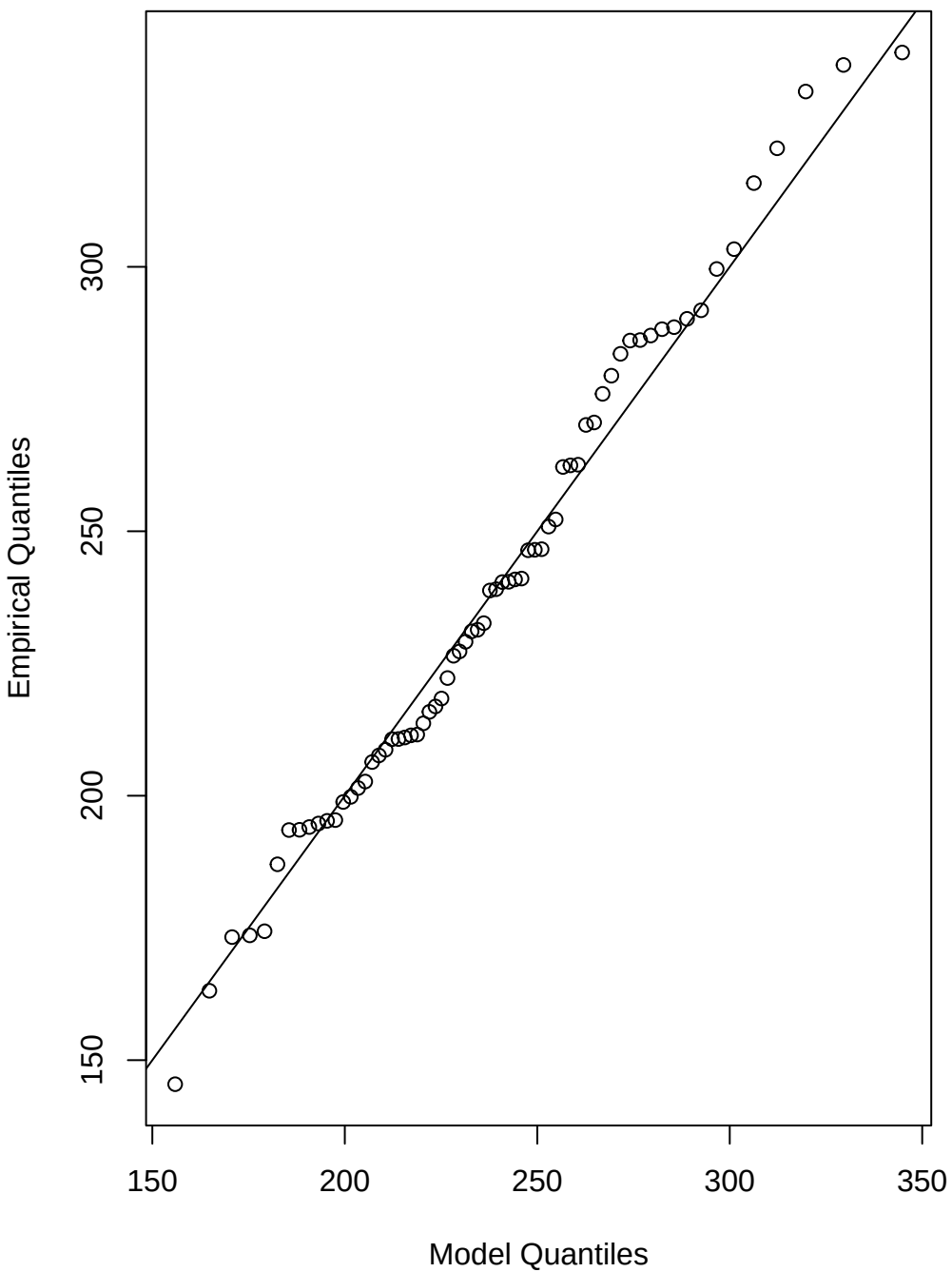
Stations 6–10: Annual Maxima vs Non-Stationary Trend (μ)



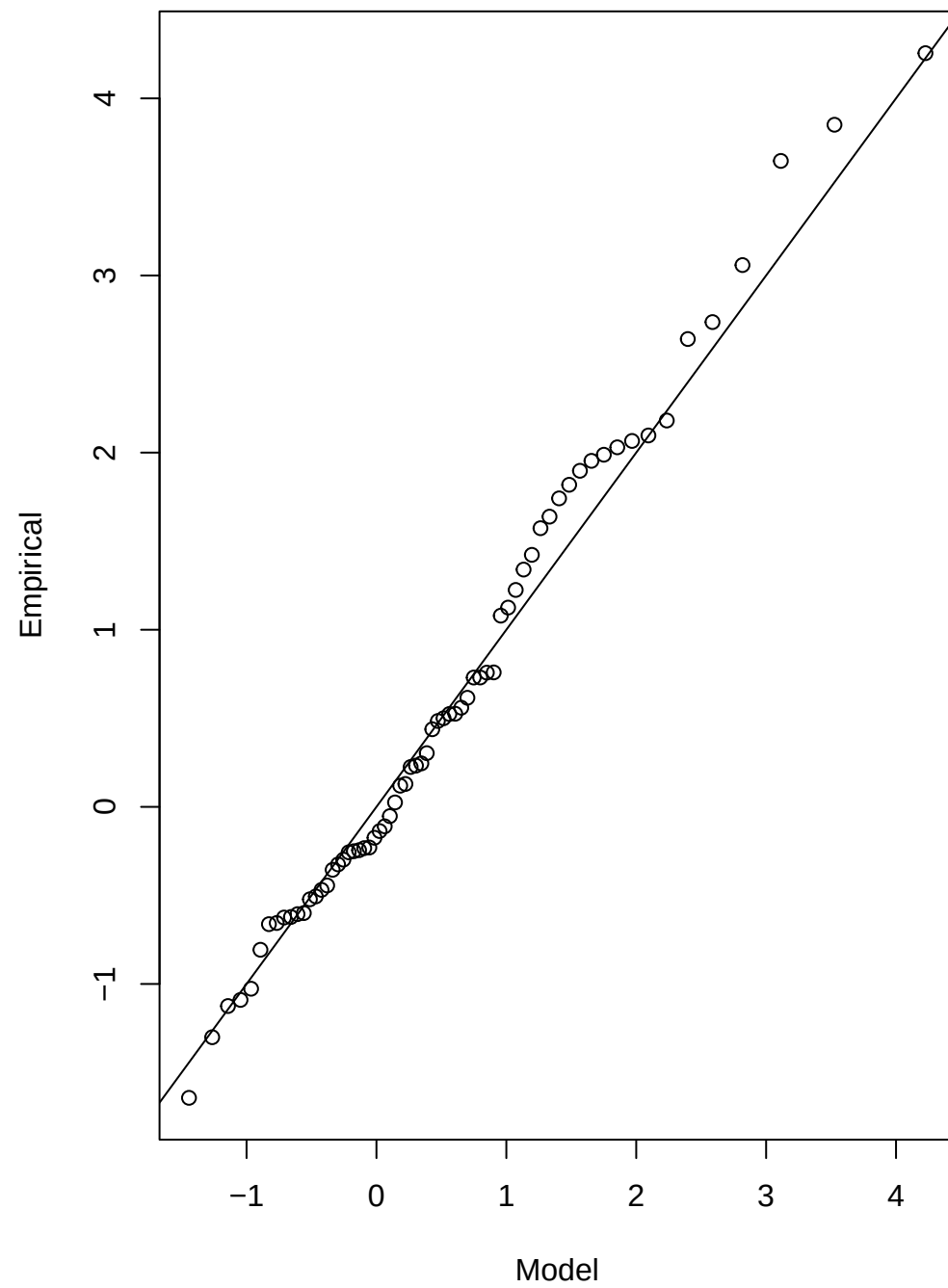
Location Parameter (μ) with 95% CIScale Parameter (σ) with 95% CIShape Parameter (ξ) with 95% CI

Diagnostic Fit (QQ) for Station: ADAMS

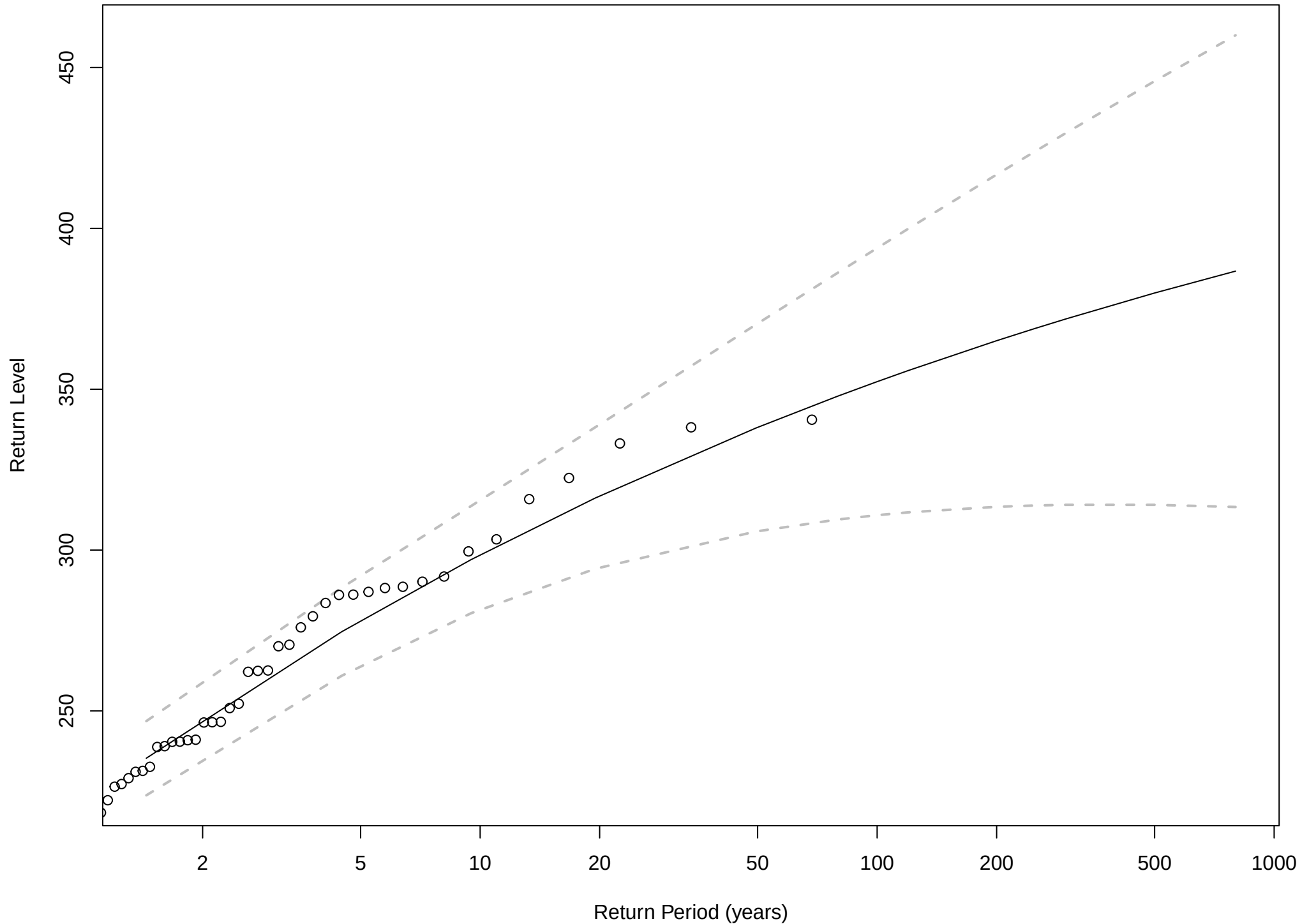
Stationary Diagnostics



Non-Stationary Diagnostics

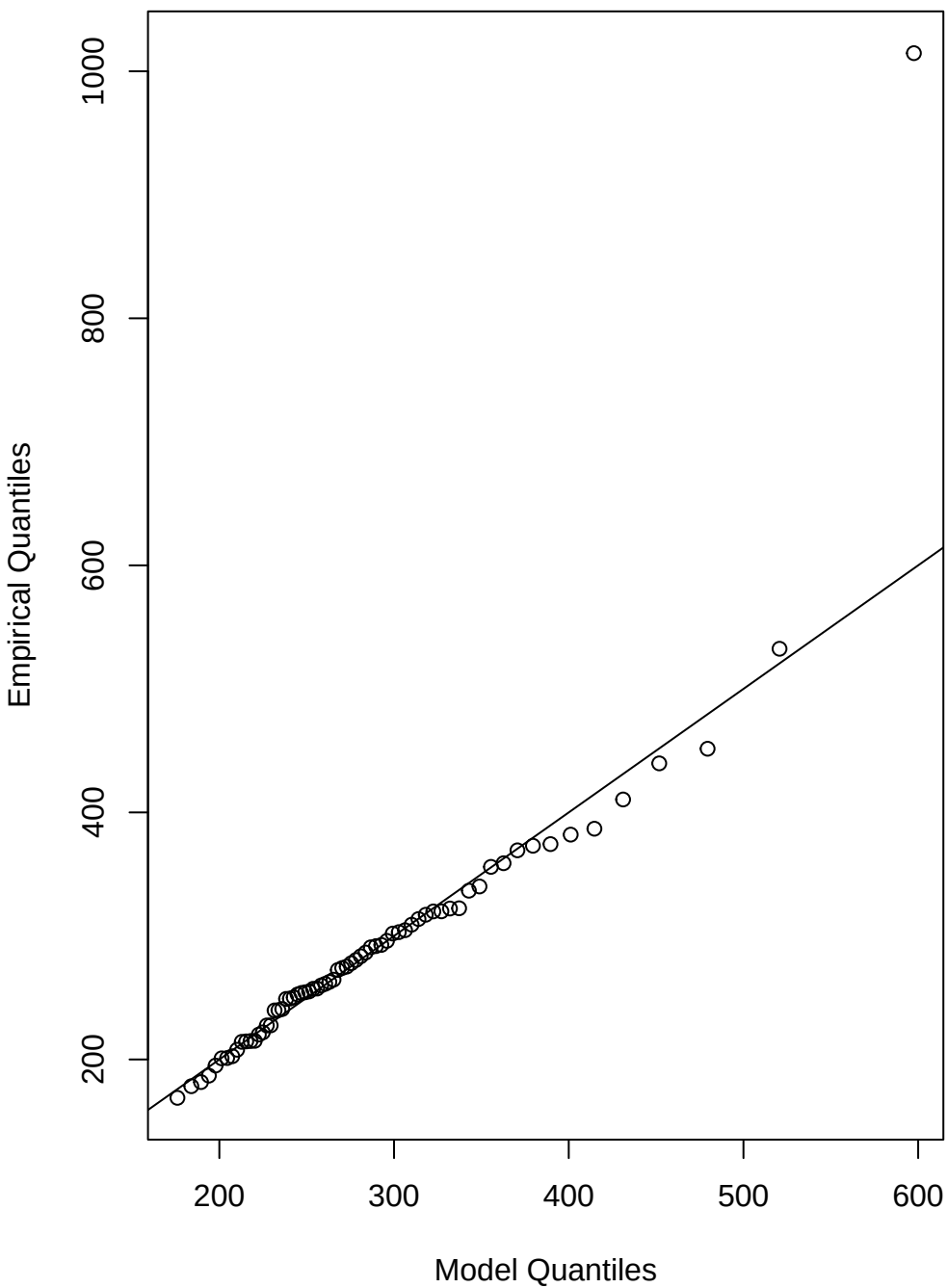


Stationary Return Level Curve: ADAMS

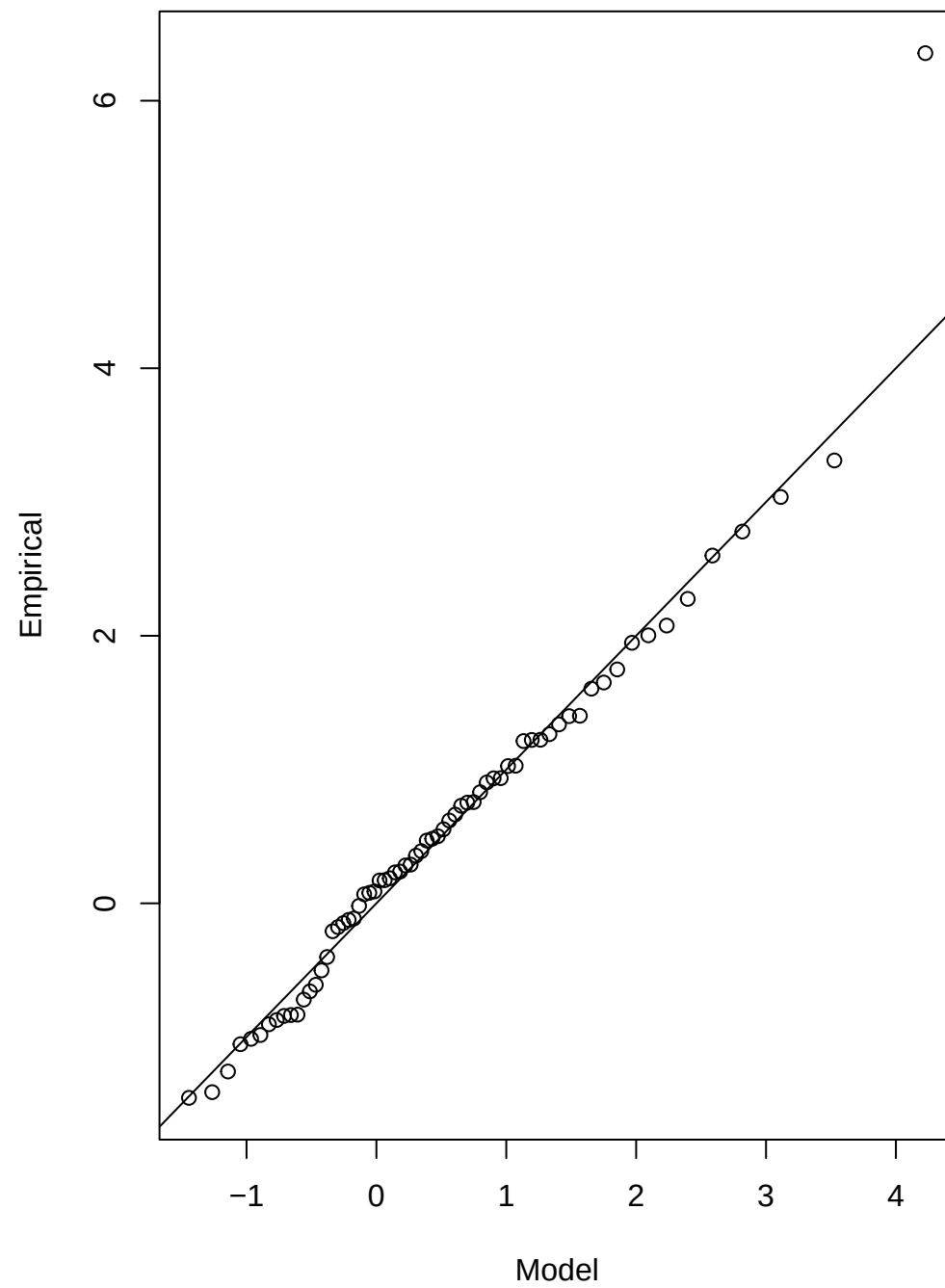


Diagnostic Fit (QQ) for Station: ARNT7

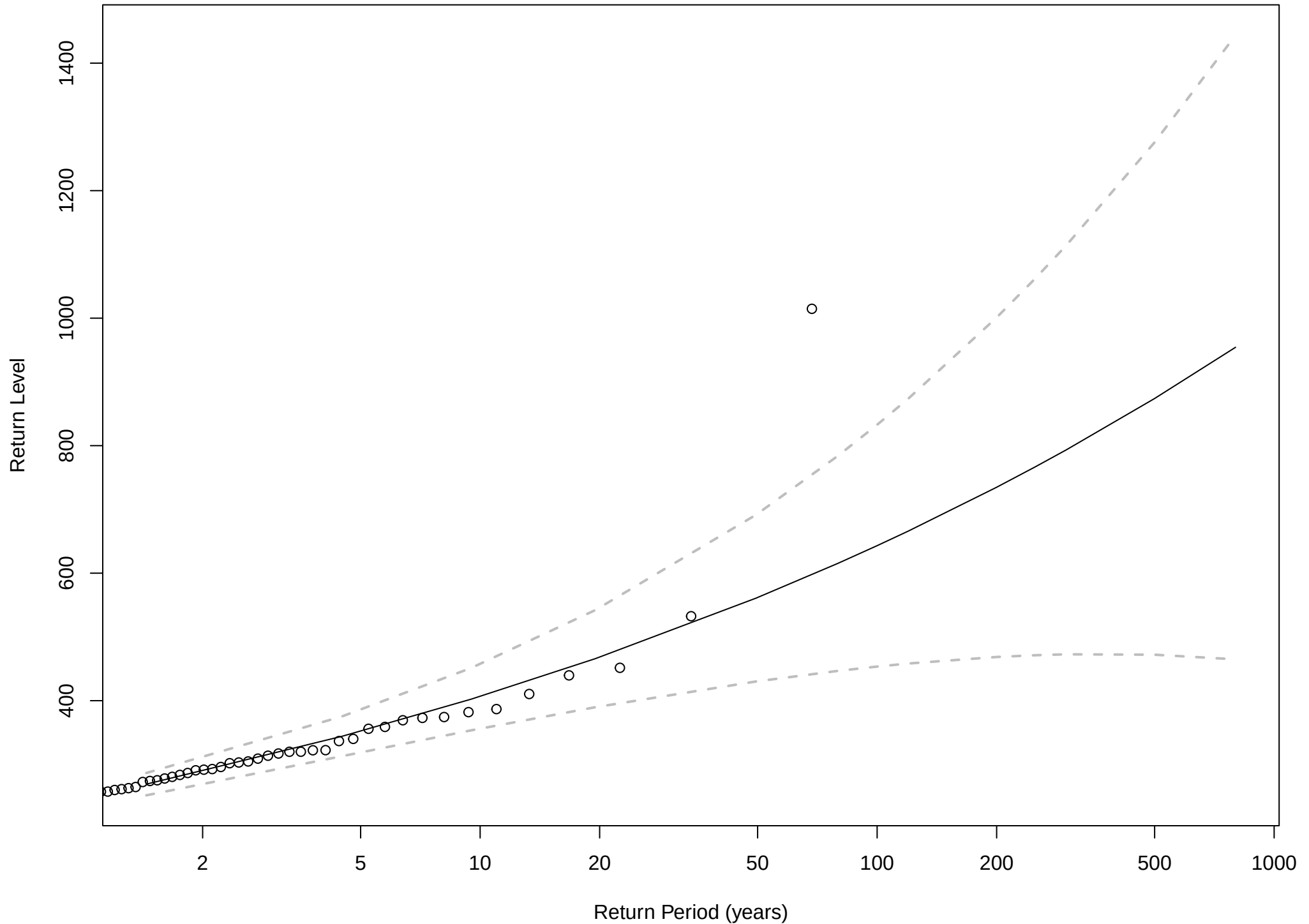
Stationary Diagnostics



Non-Stationary Diagnostics

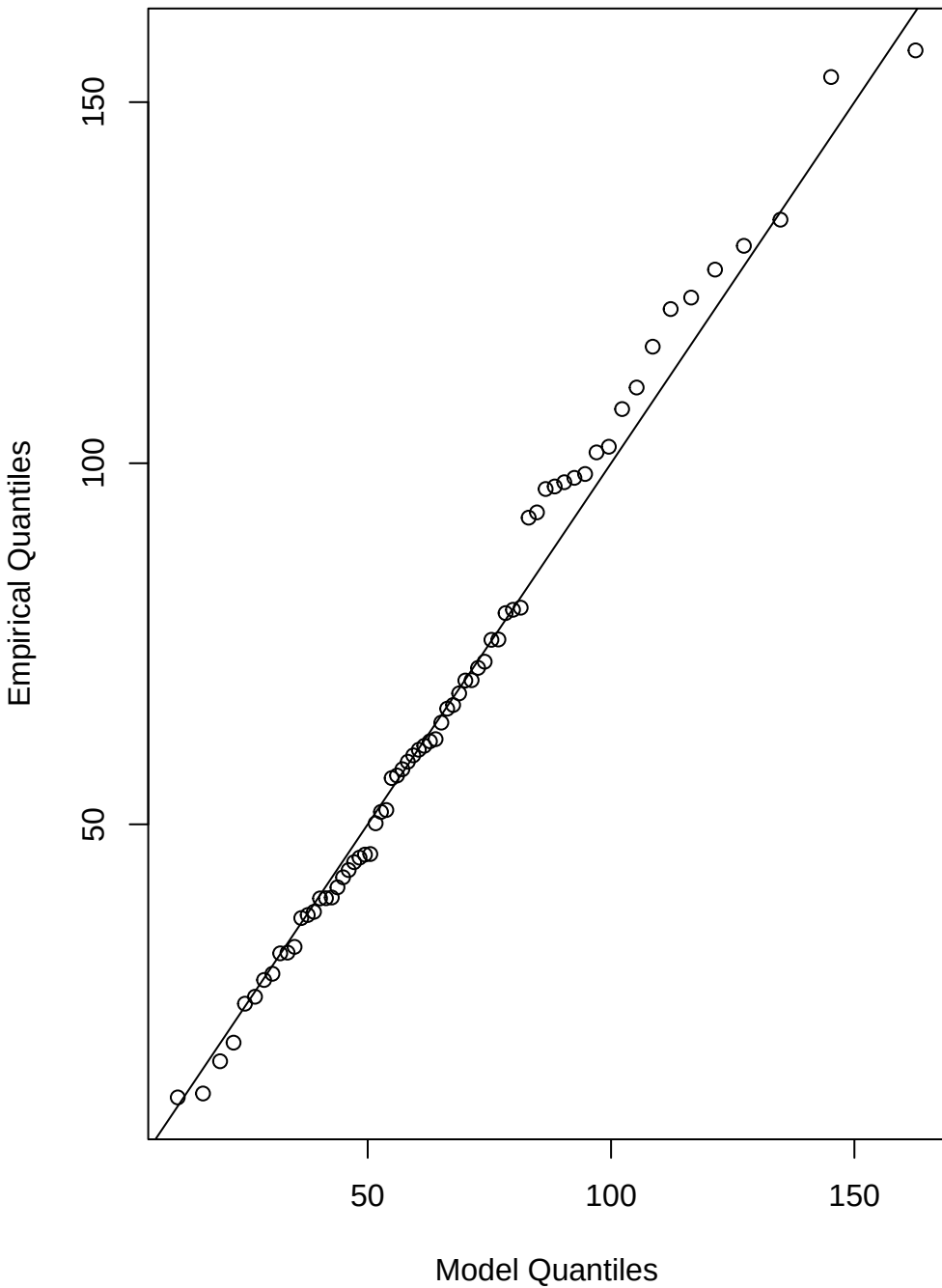


Stationary Return Level Curve: ARNT7

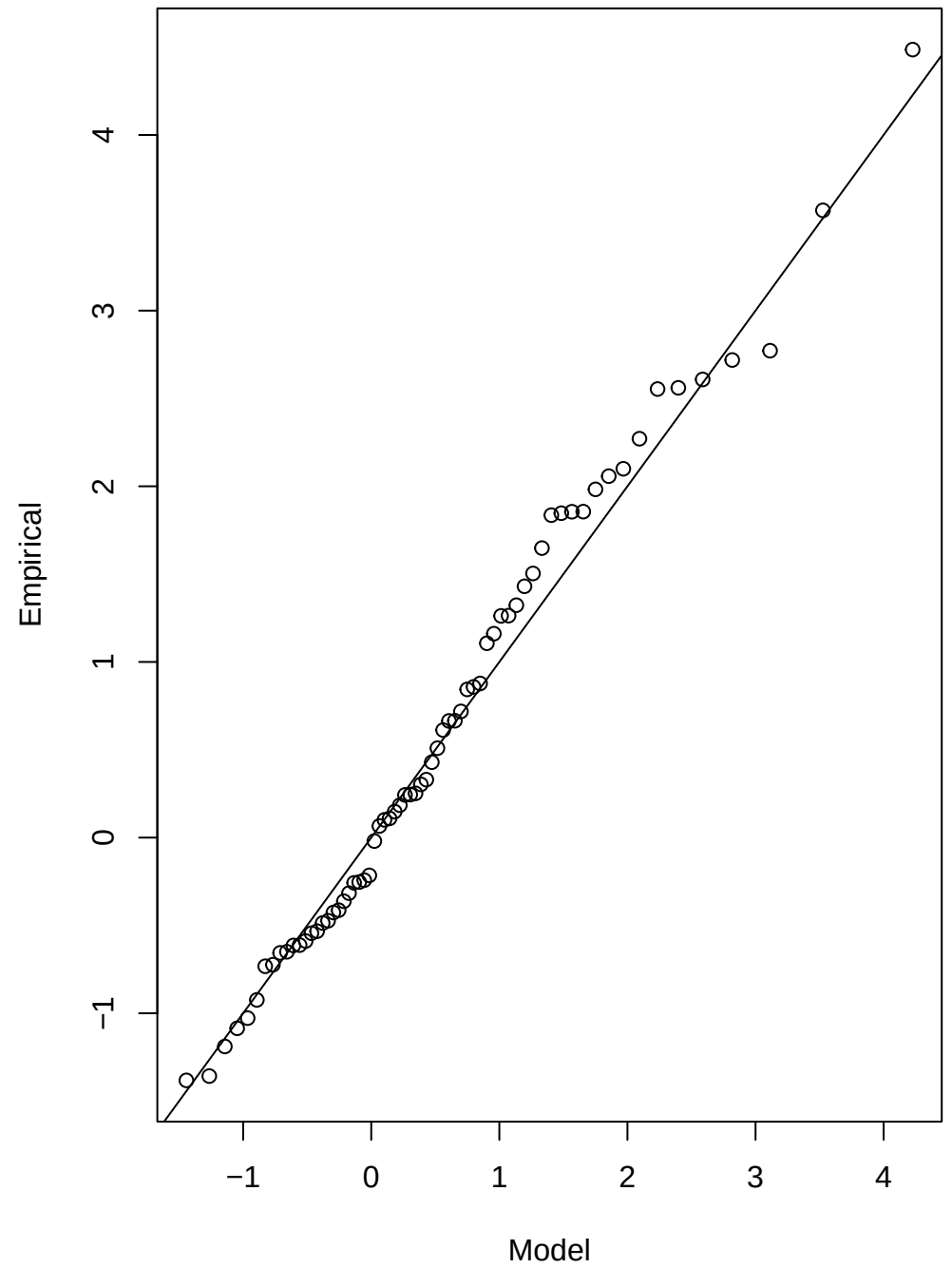


Diagnostic Fit (QQ) for Station: ASHNO

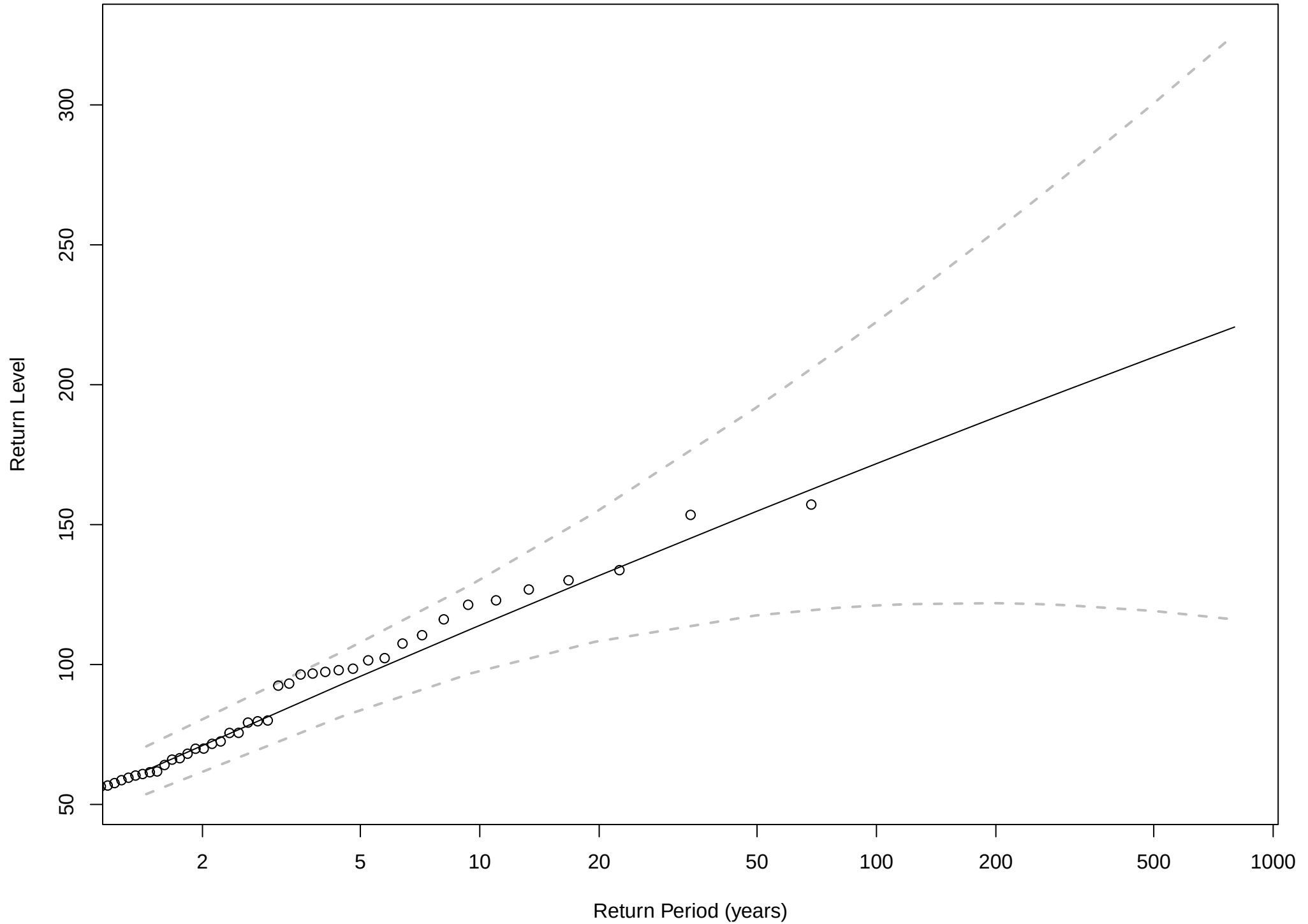
Stationary Diagnostics



Non-Stationary Diagnostics

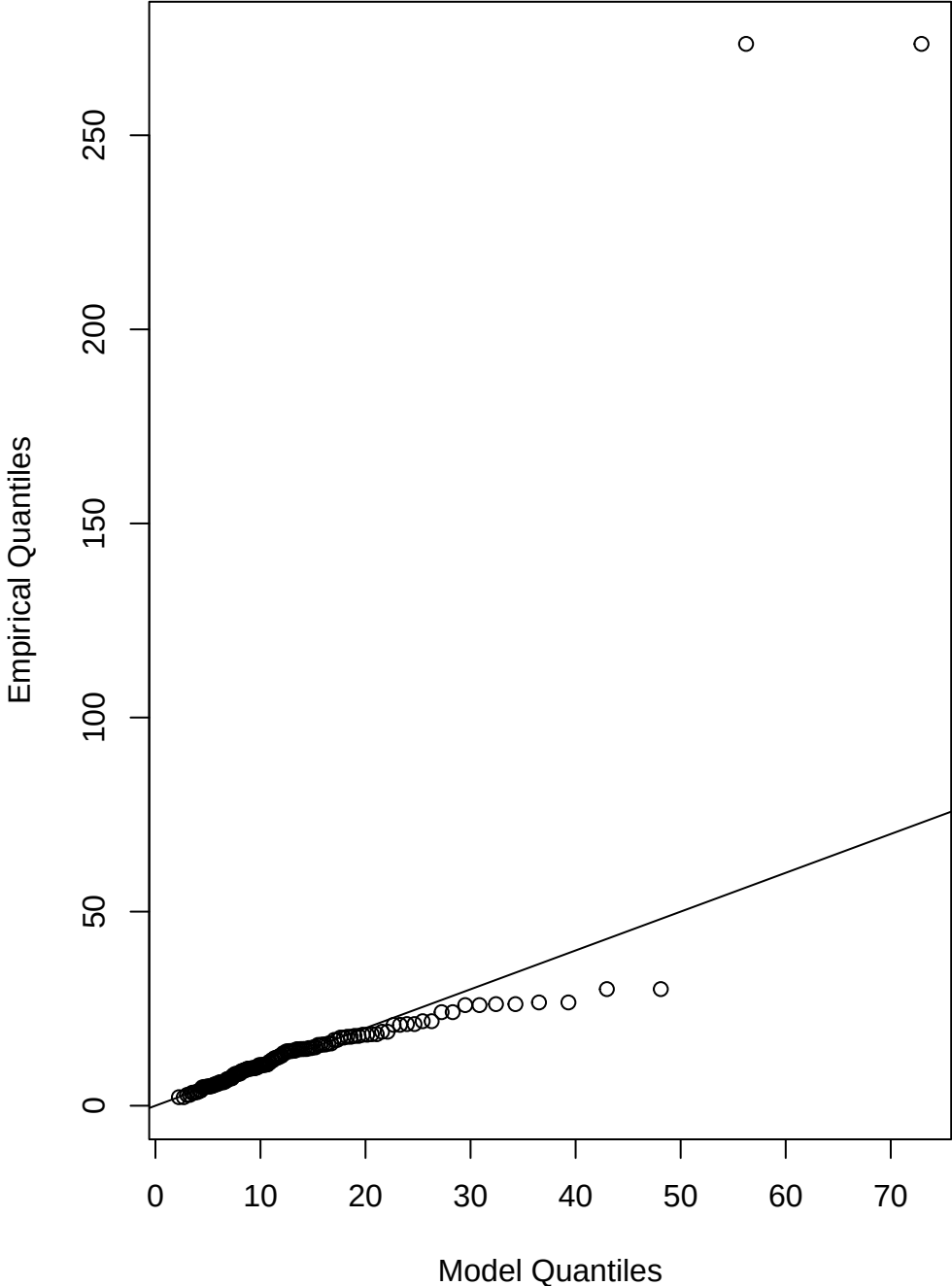


Stationary Return Level Curve: ASHNO

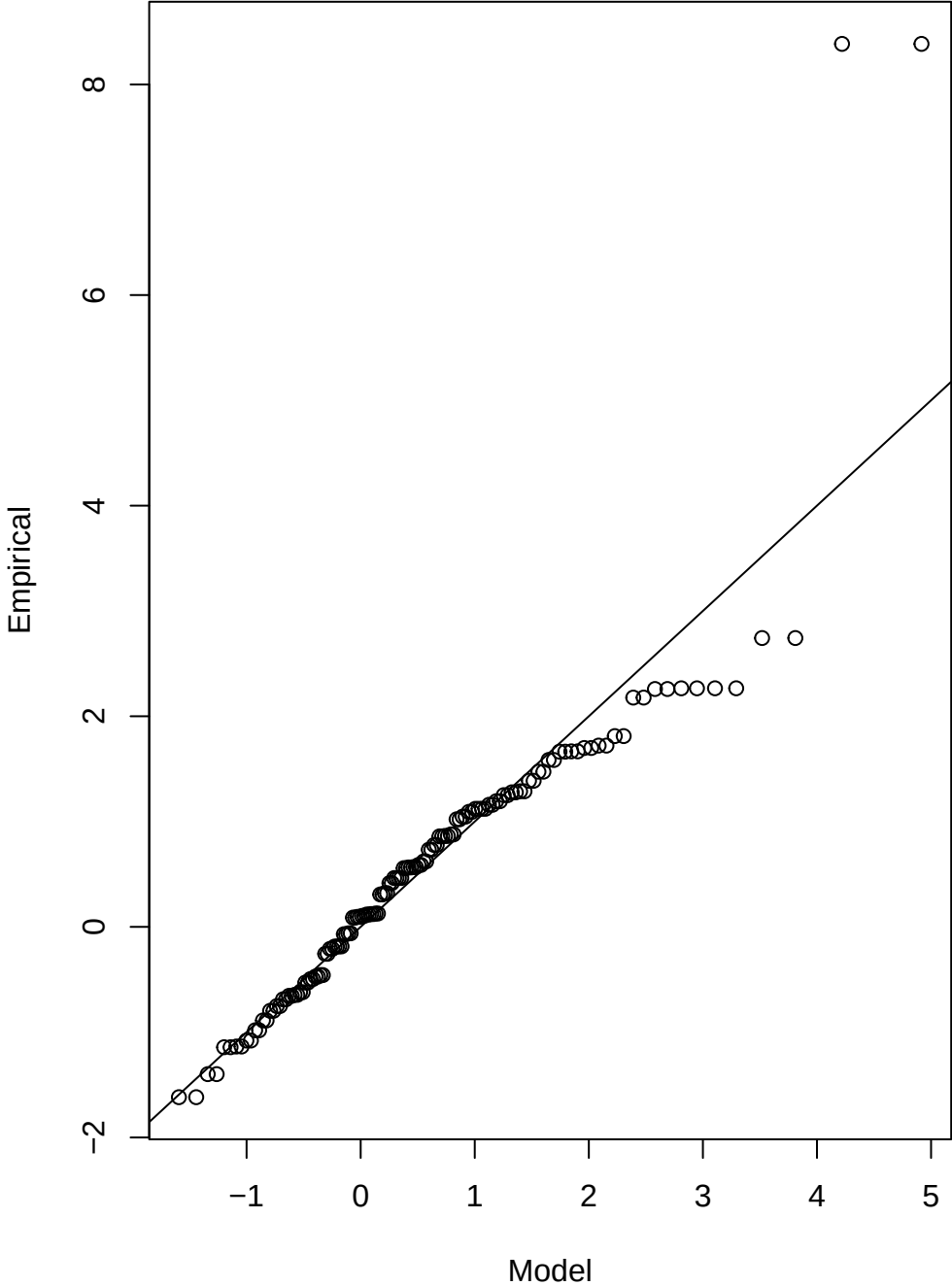


Diagnostic Fit (QQ) for Station: BAEZA

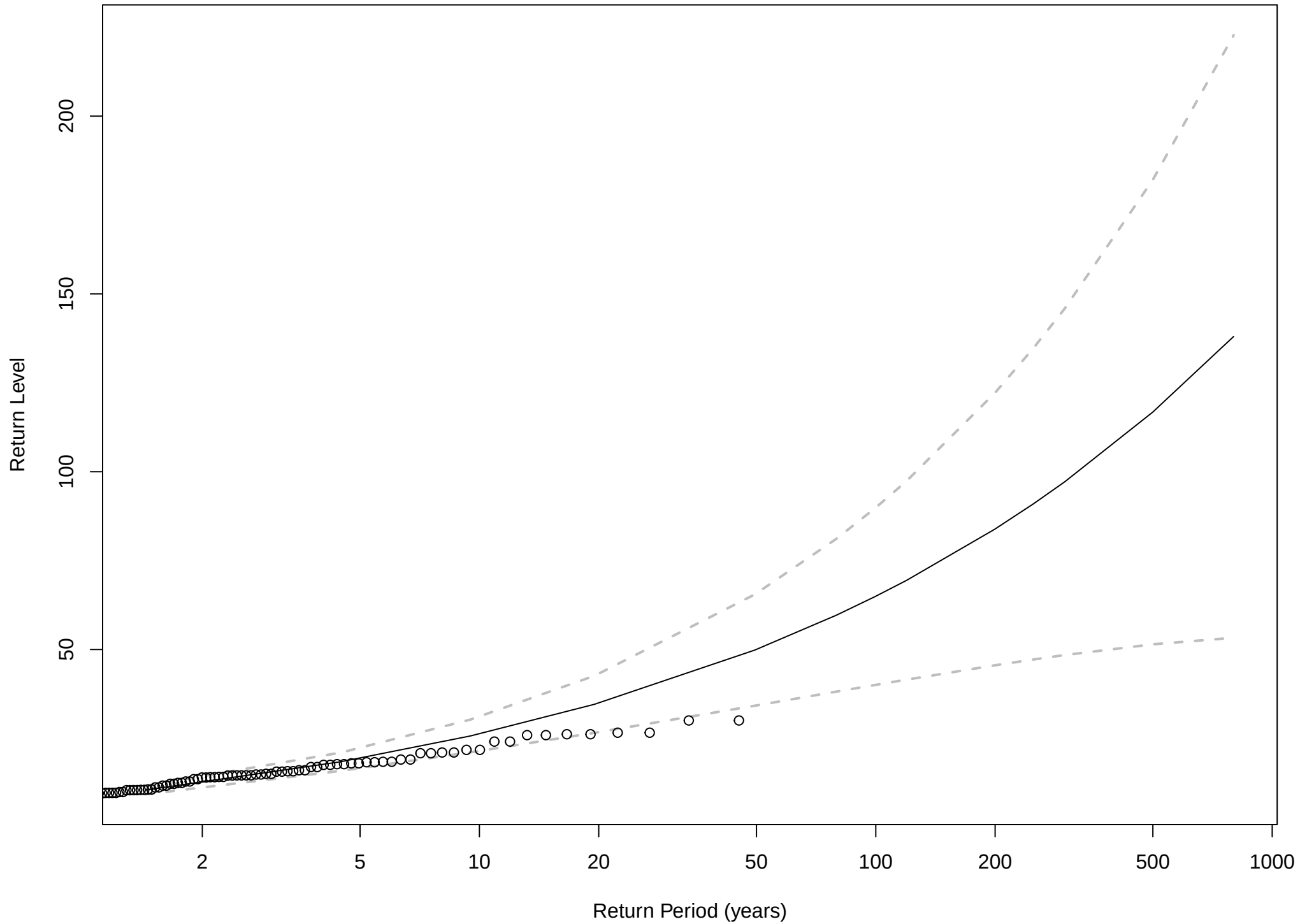
Stationary Diagnostics



Non-Stationary Diagnostics

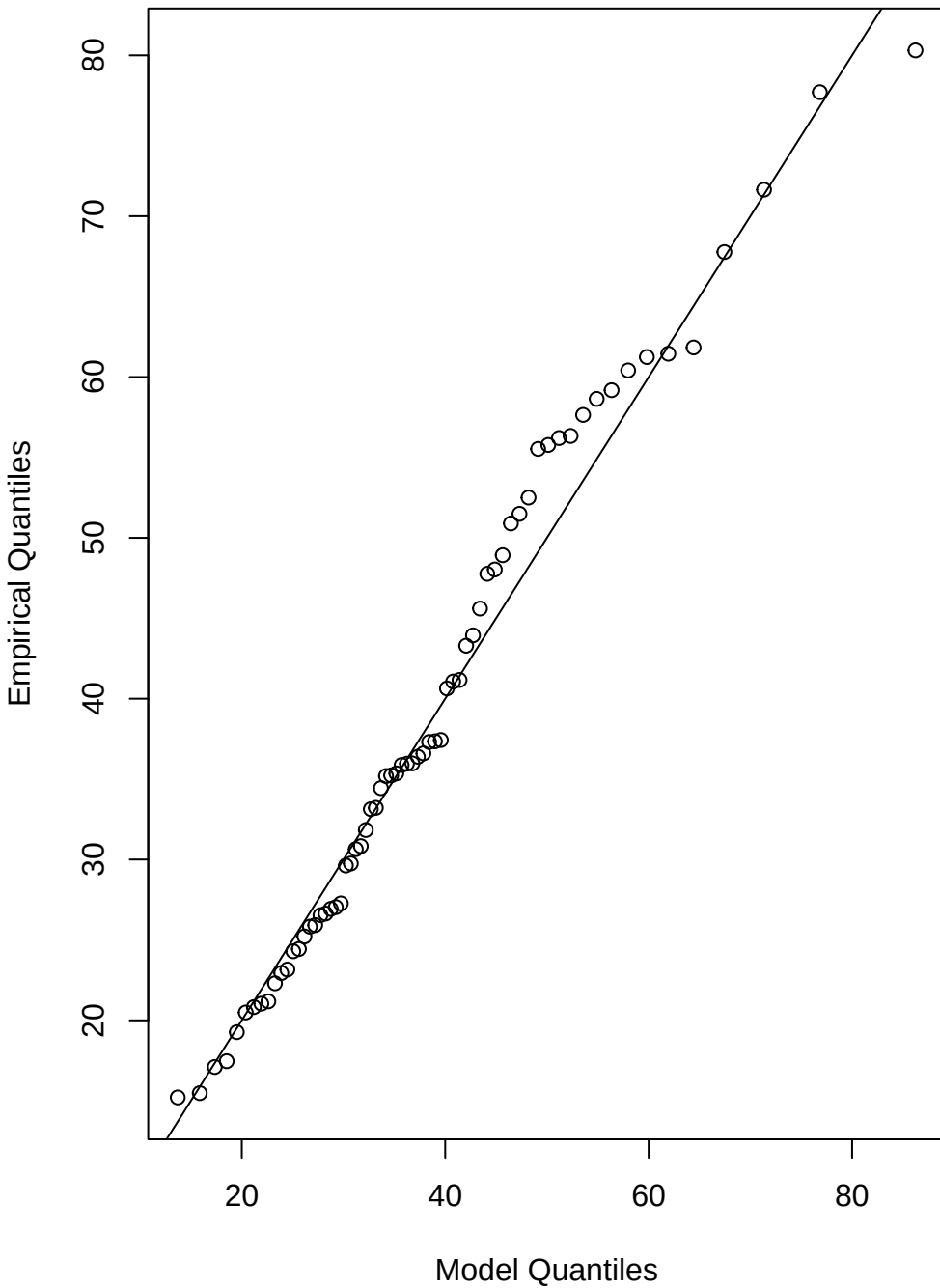


Stationary Return Level Curve: BAEZA

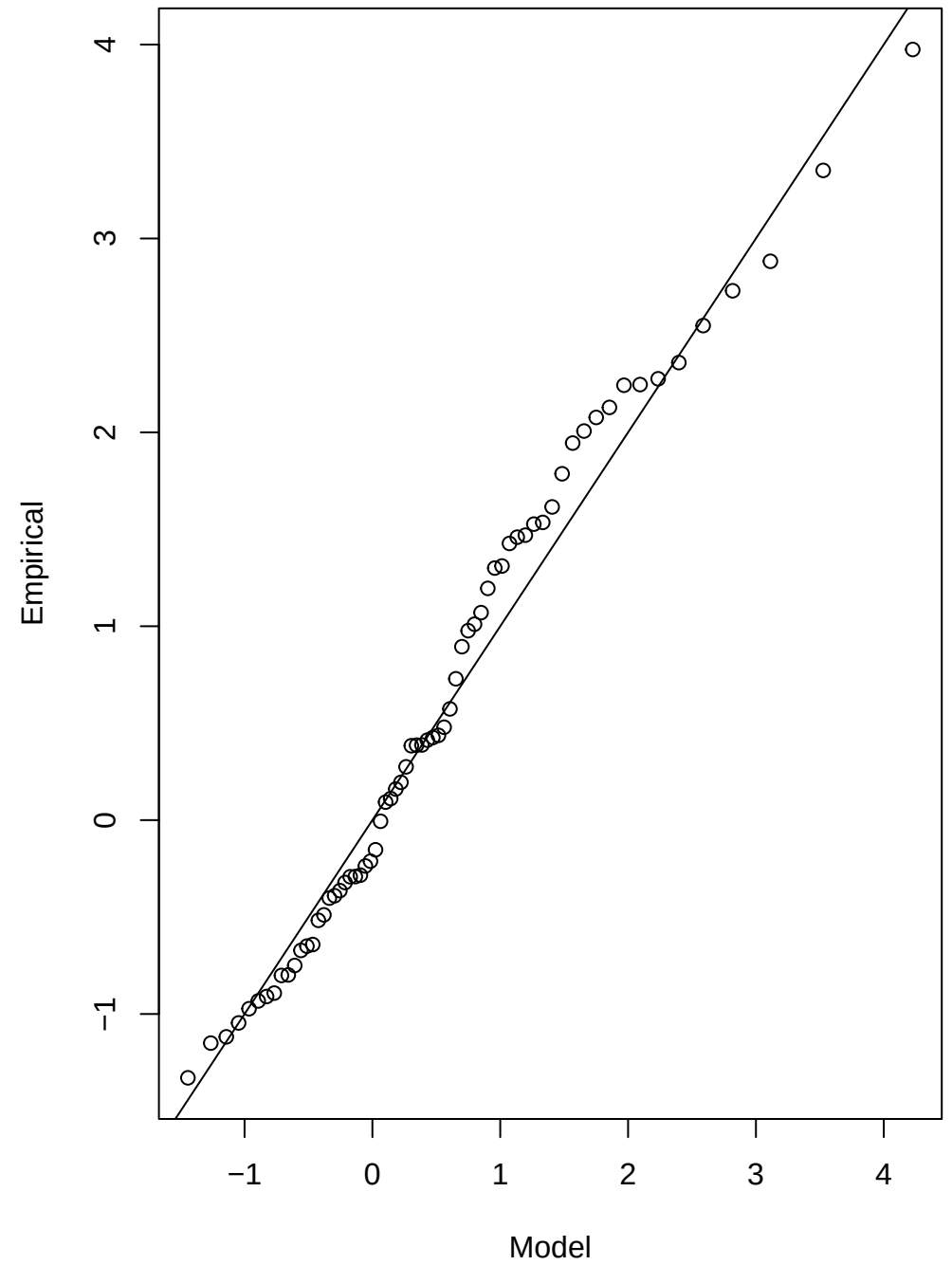


Diagnostic Fit (QQ) for Station: BAKER

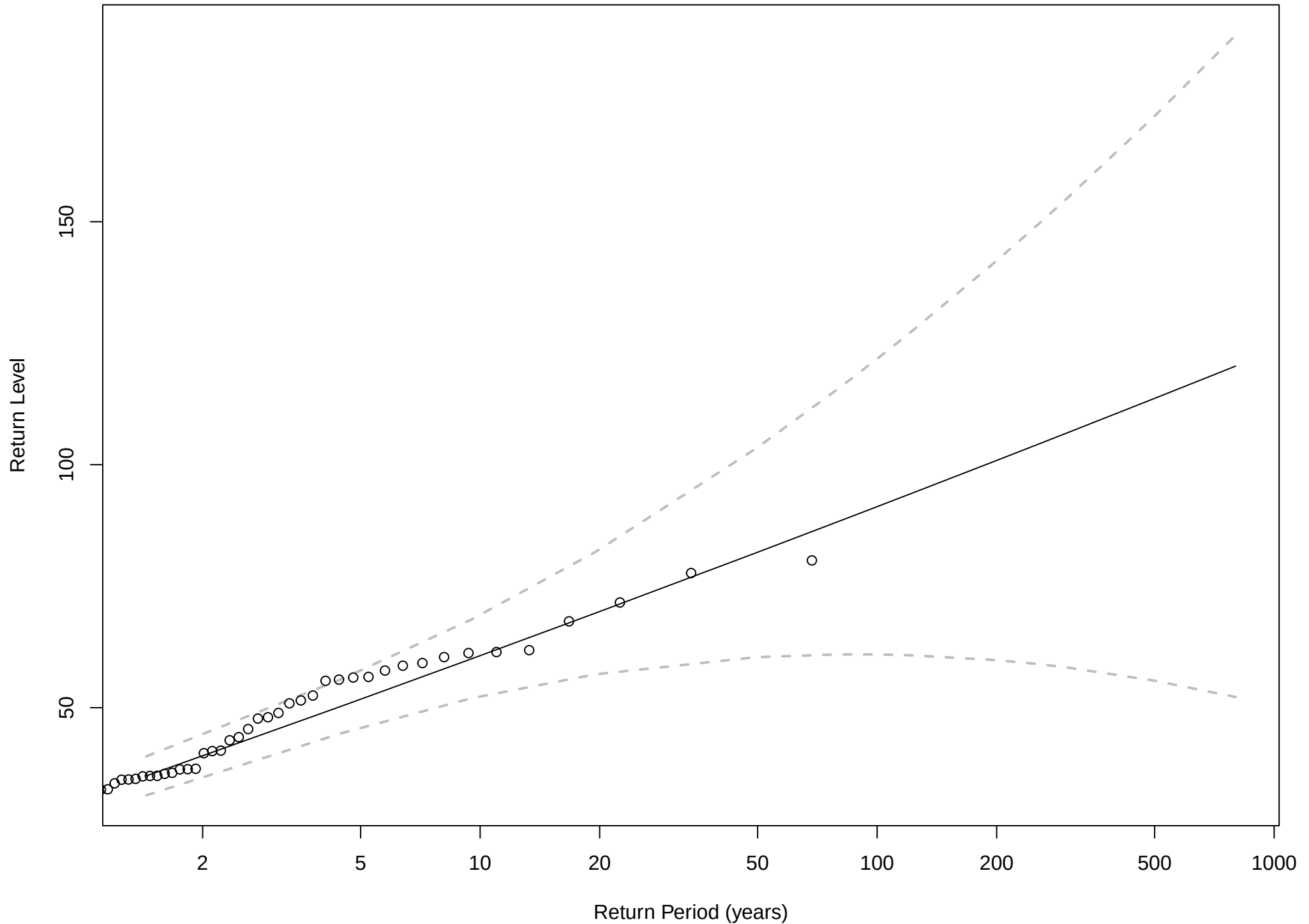
Stationary Diagnostics



Non-Stationary Diagnostics

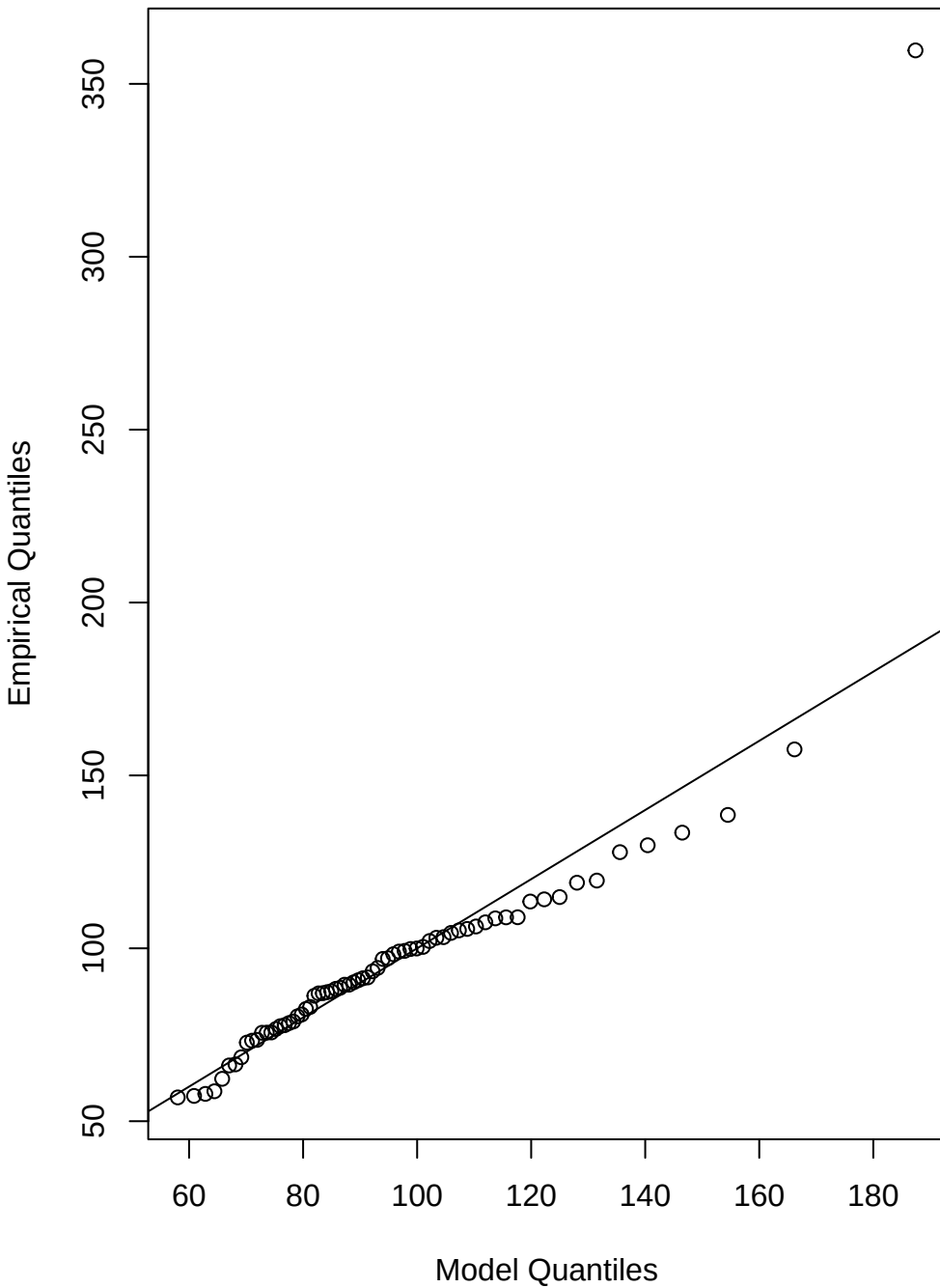


Stationary Return Level Curve: BAKER

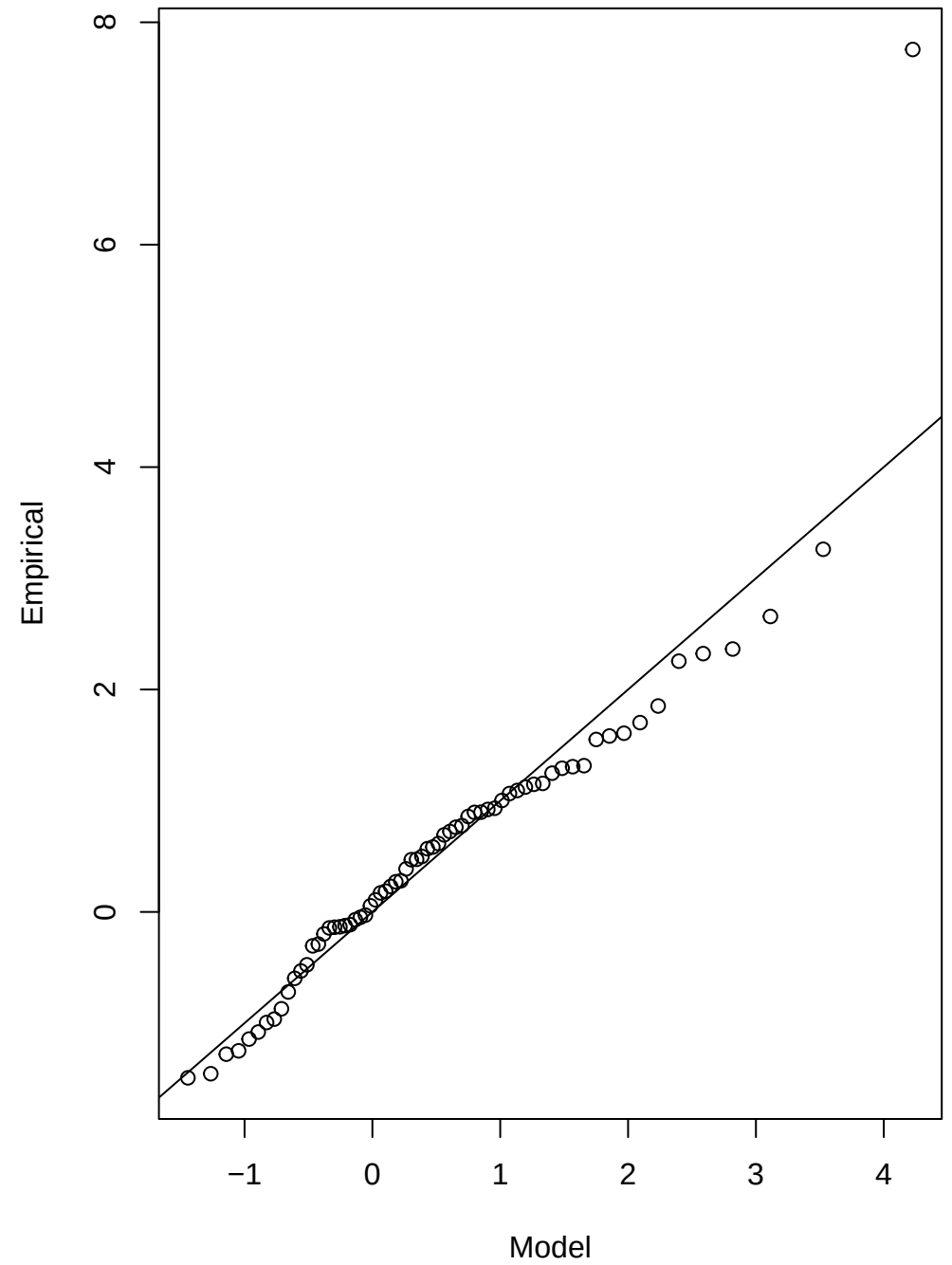


Diagnostic Fit (QQ) for Station: BARRM

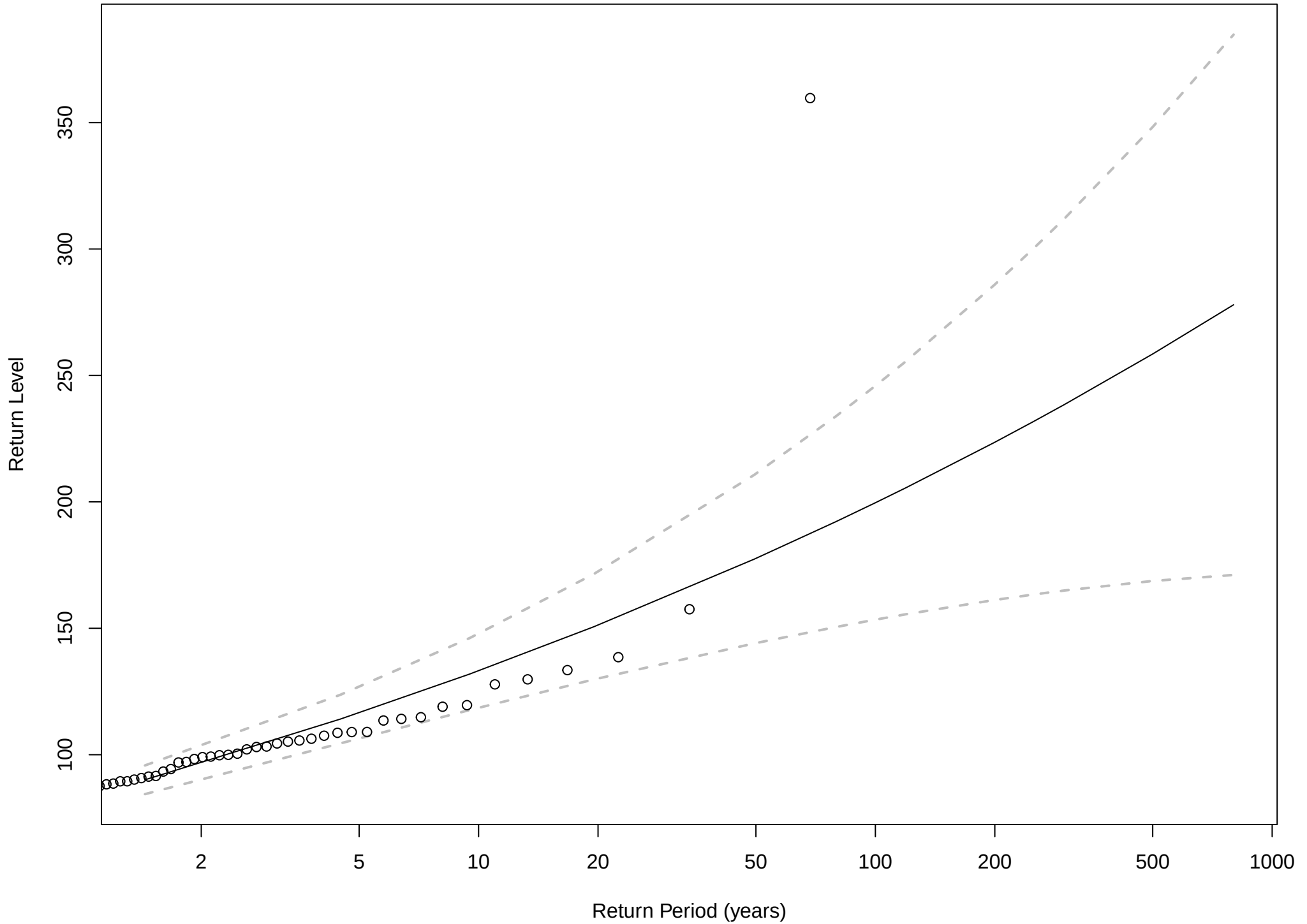
Stationary Diagnostics



Non-Stationary Diagnostics

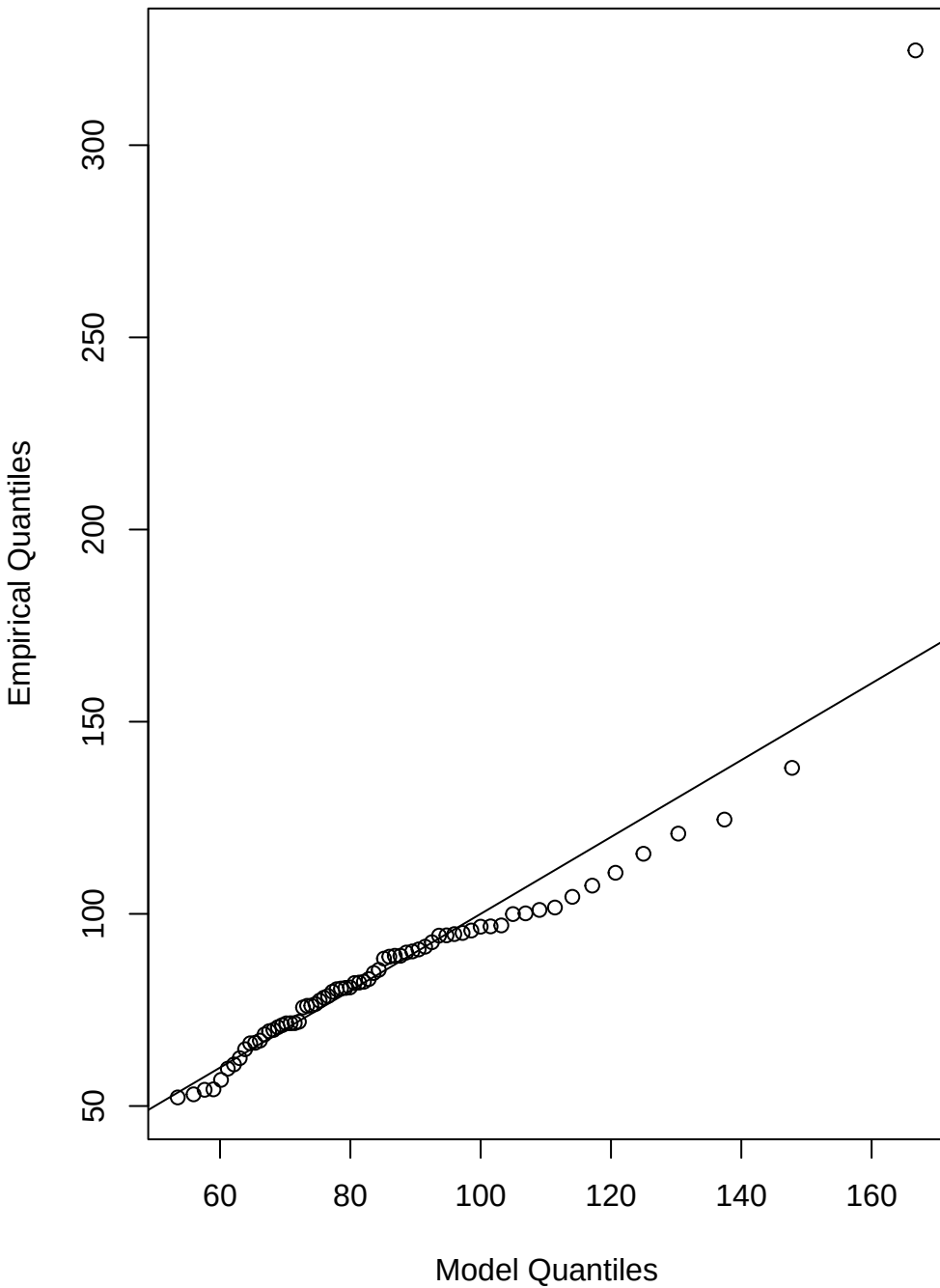


Stationary Return Level Curve: BARRM

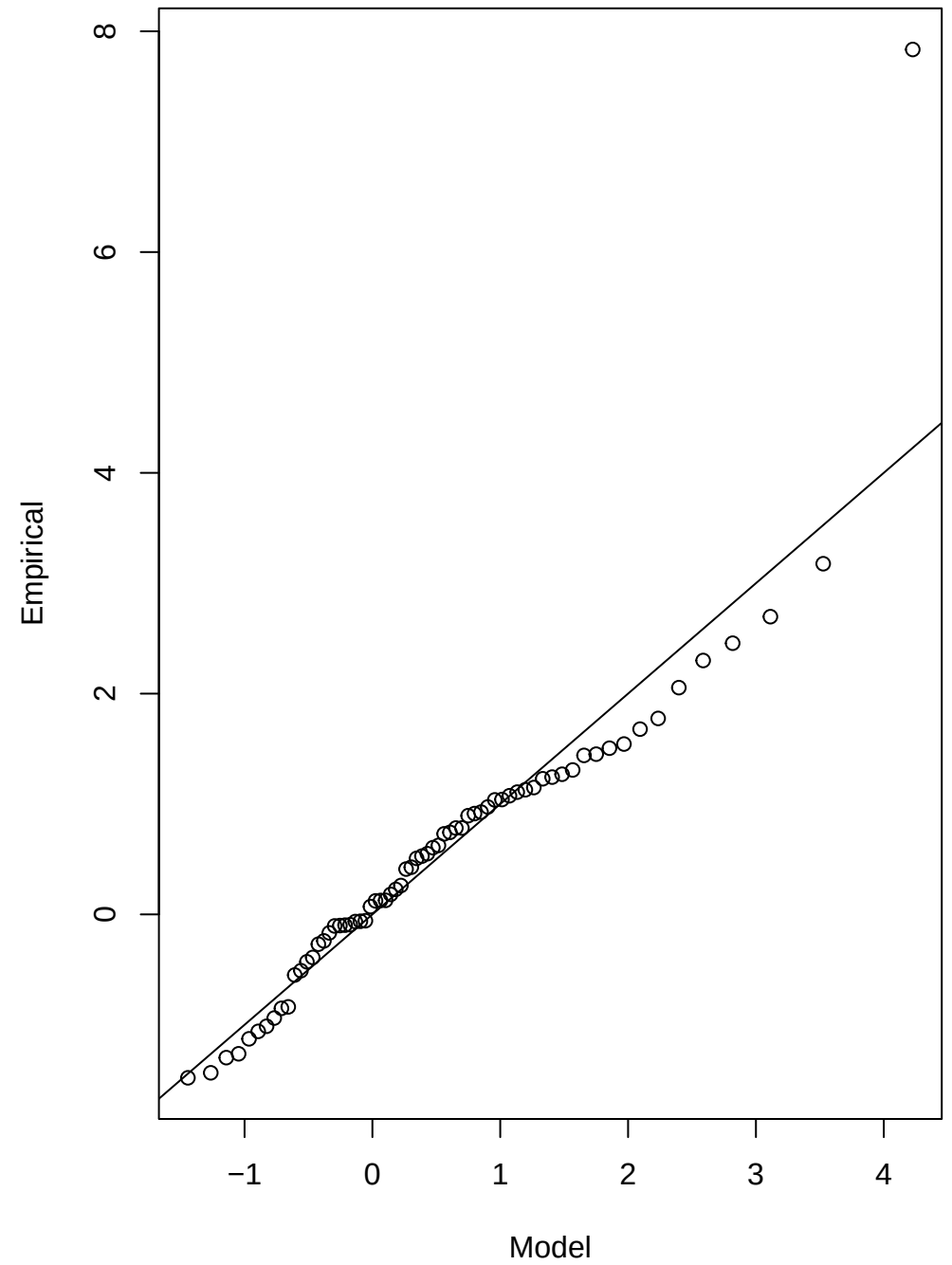


Diagnostic Fit (QQ) for Station: BARRS

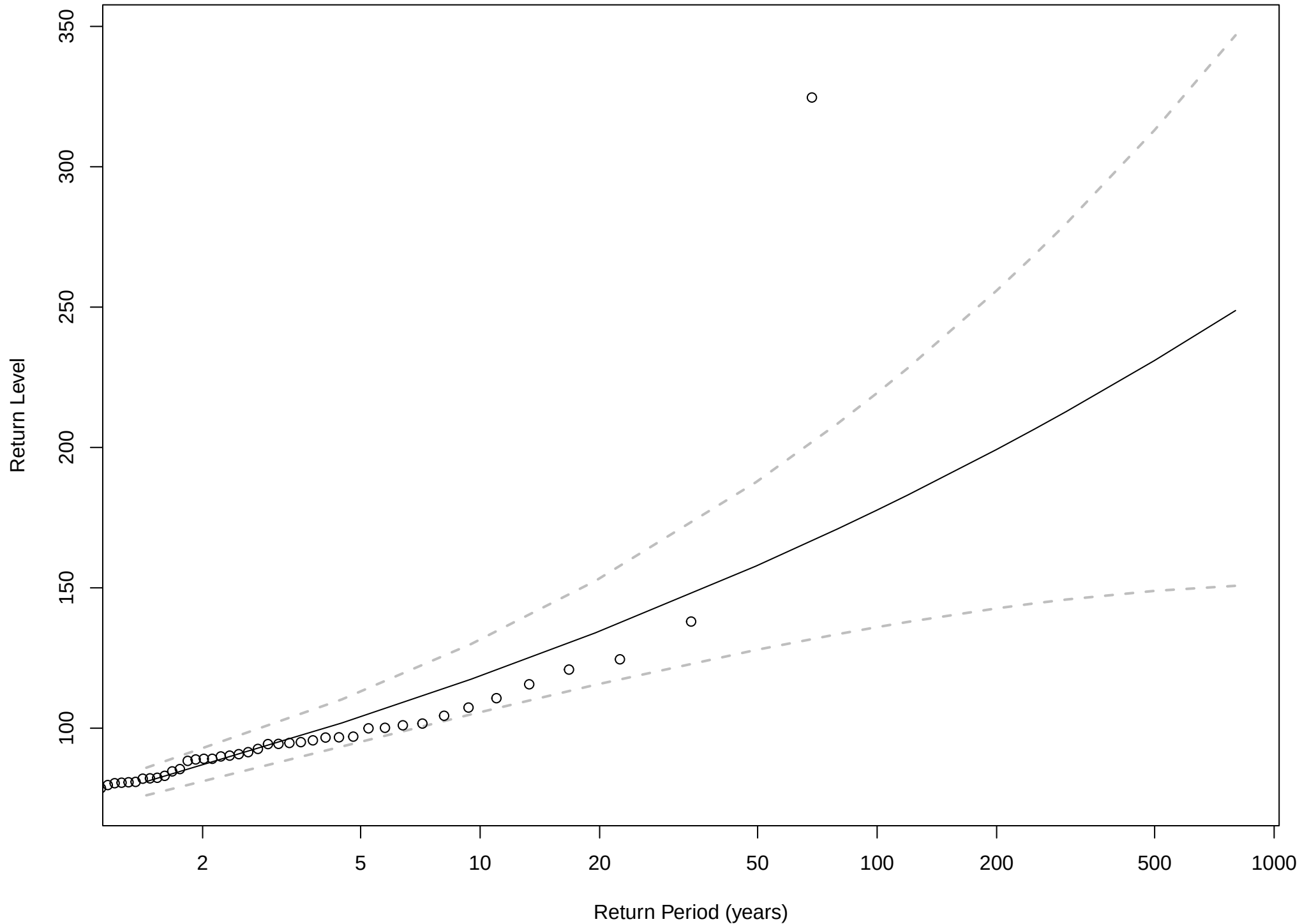
Stationary Diagnostics



Non-Stationary Diagnostics

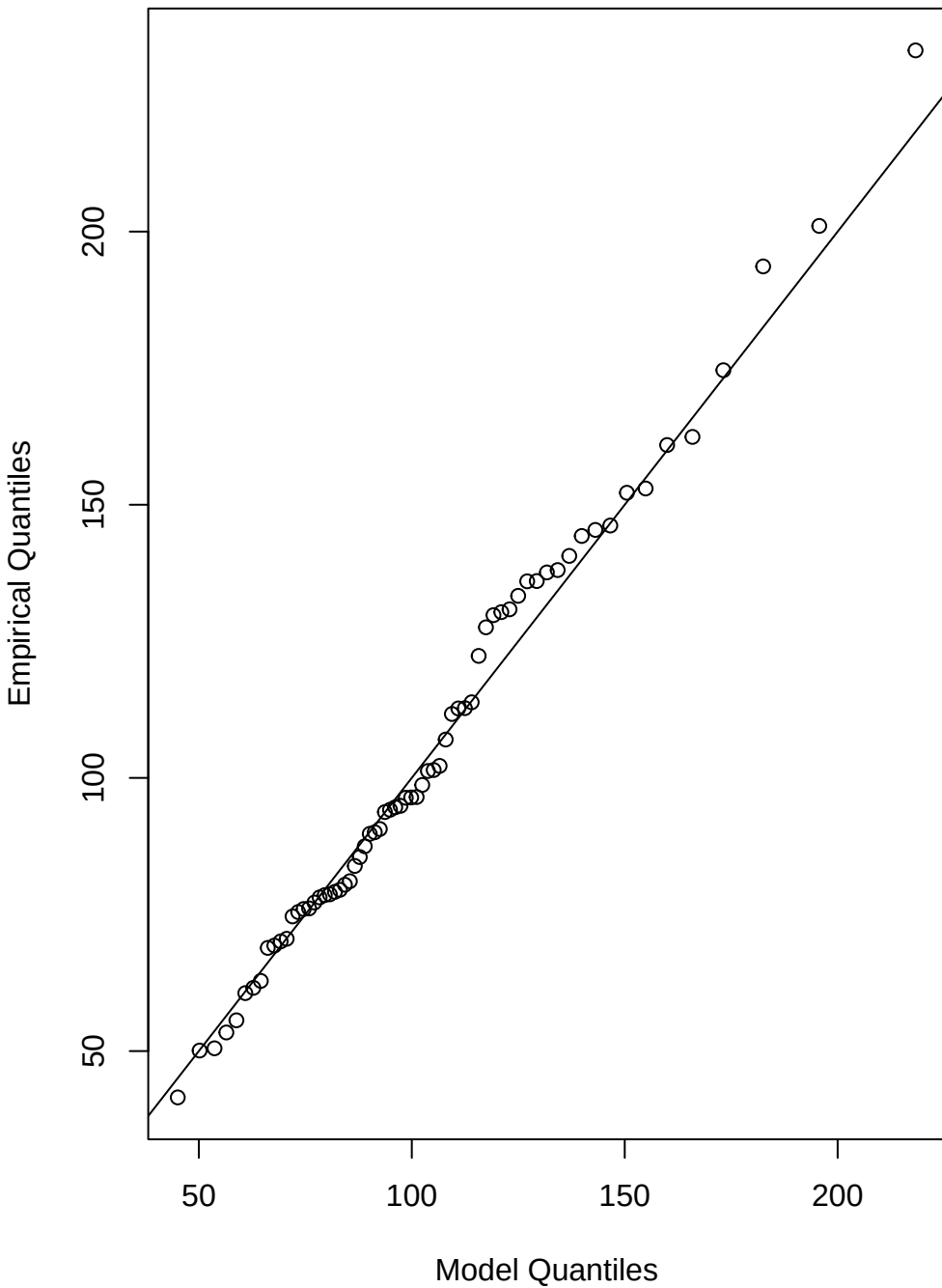


Stationary Return Level Curve: BARRS

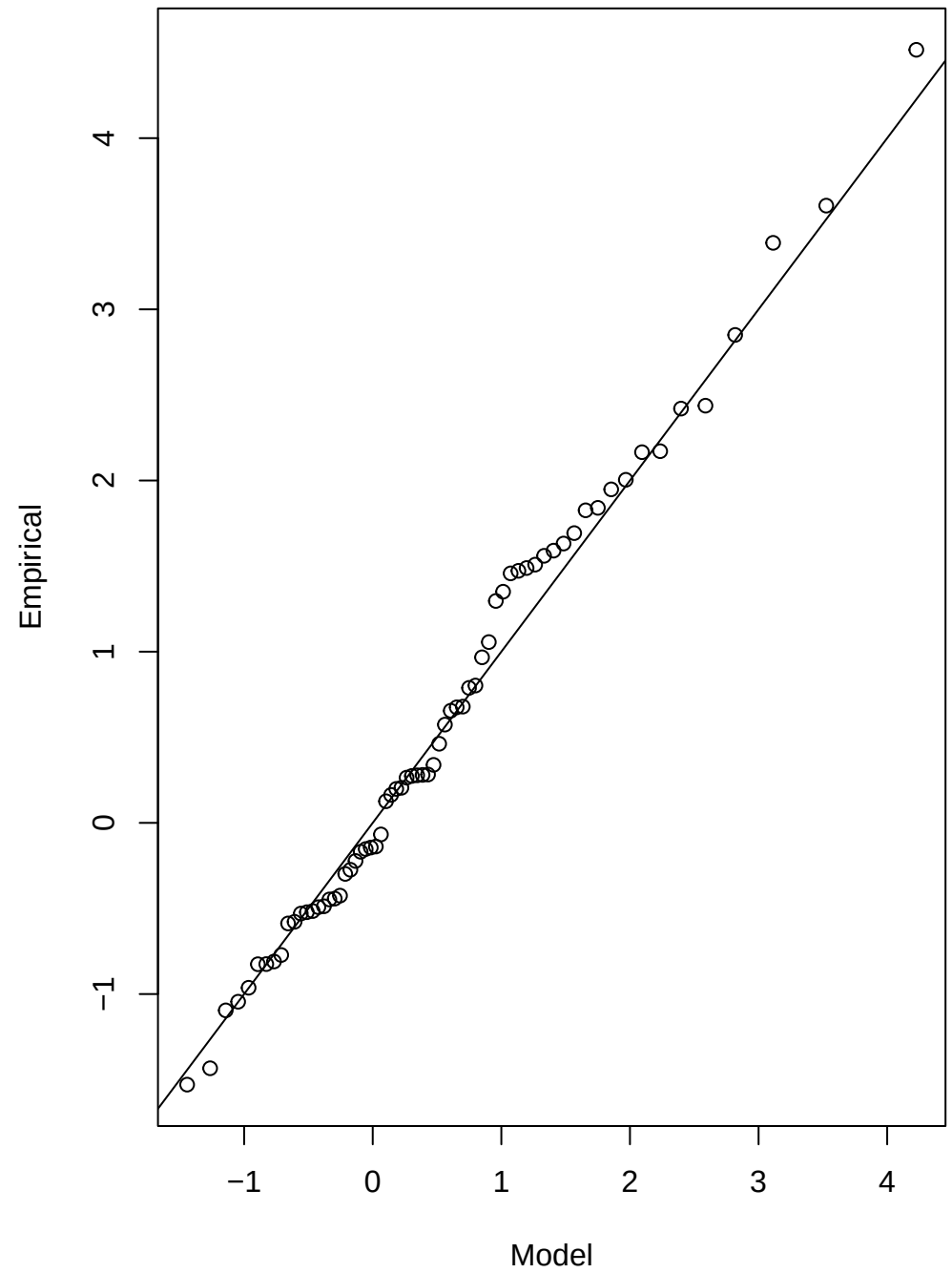


Diagnostic Fit (QQ) for Station: BCHAL

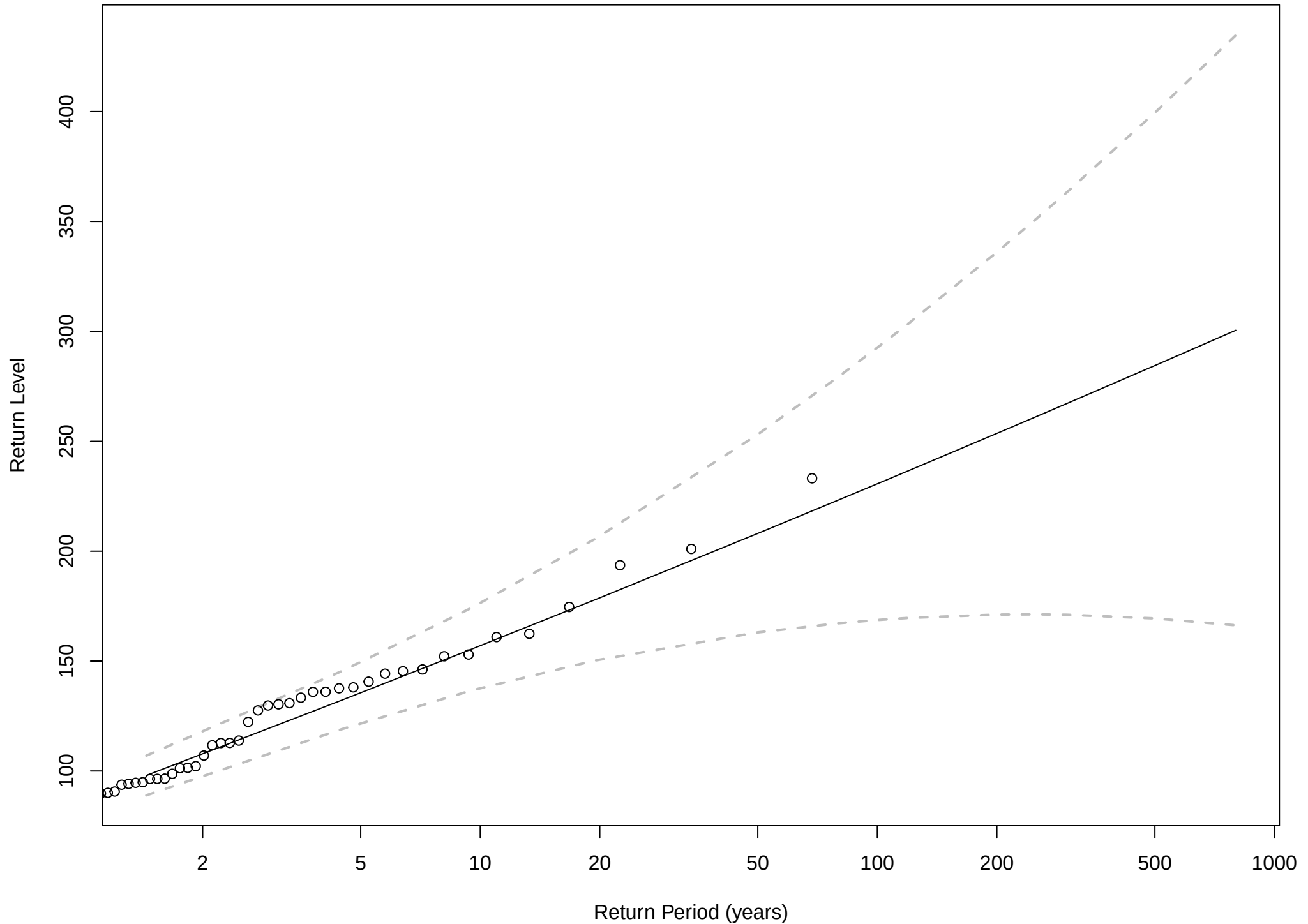
Stationary Diagnostics



Non-Stationary Diagnostics

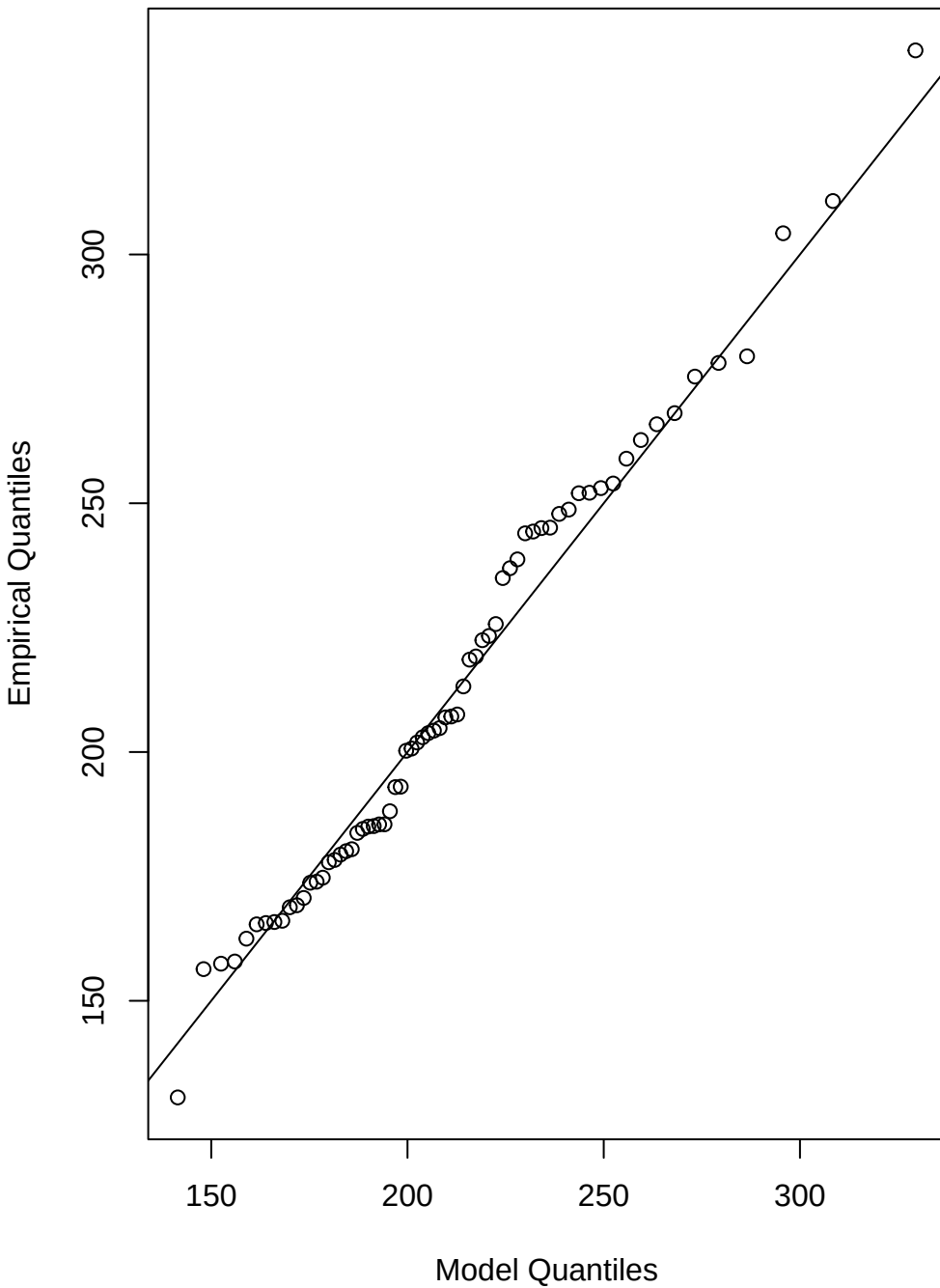


Stationary Return Level Curve: BCHAL

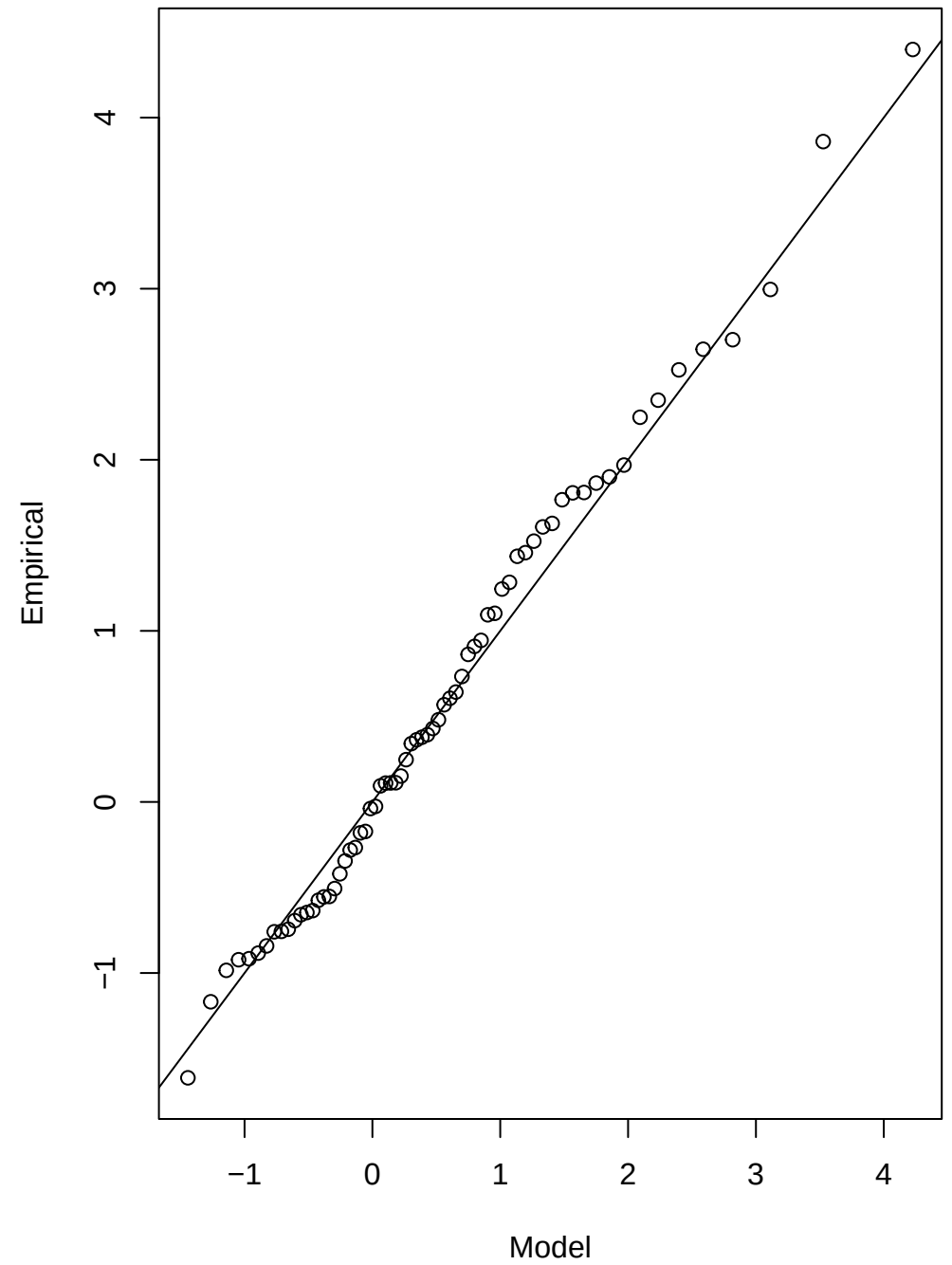


Diagnostic Fit (QQ) for Station: BCHLJ

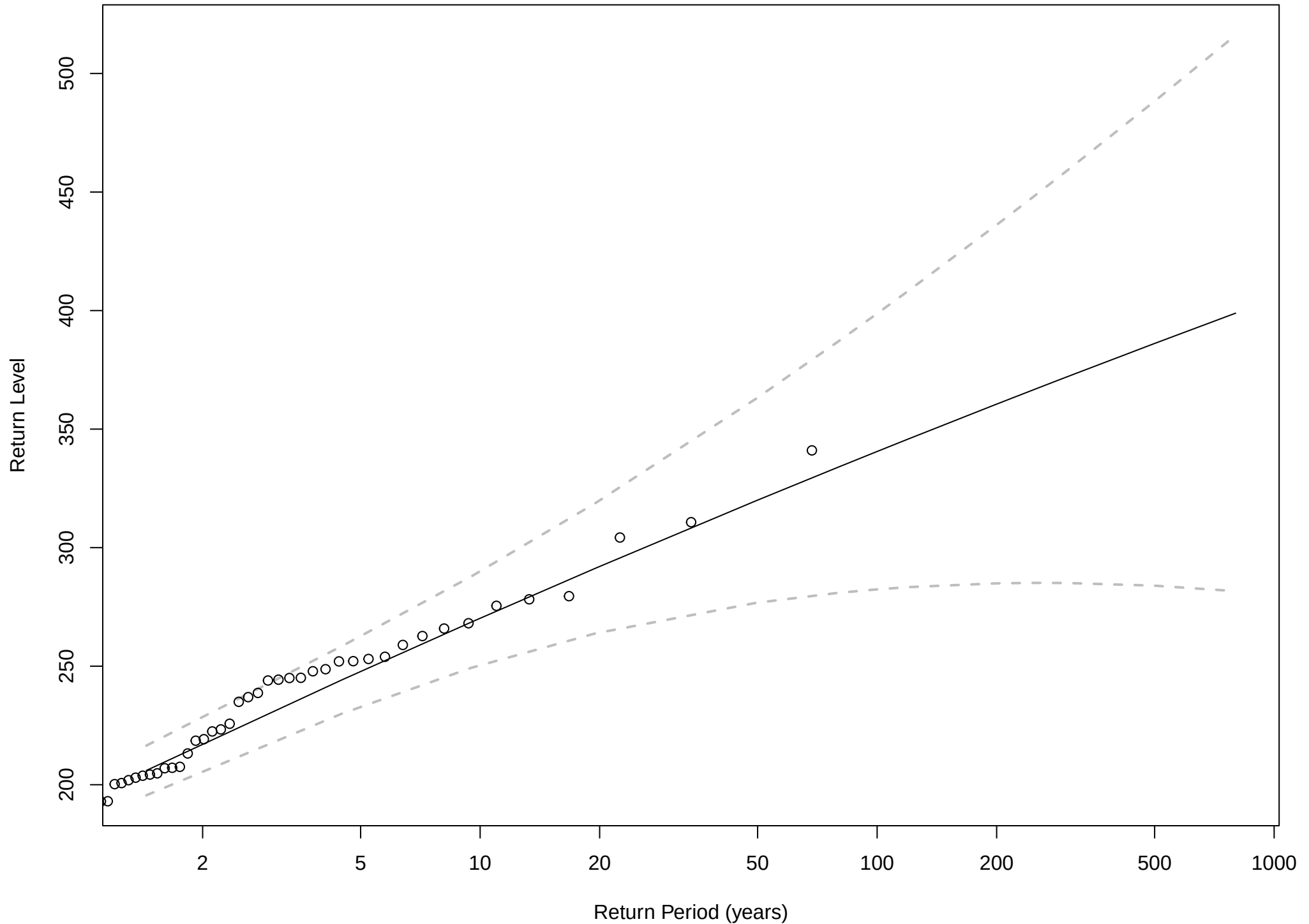
Stationary Diagnostics



Non-Stationary Diagnostics

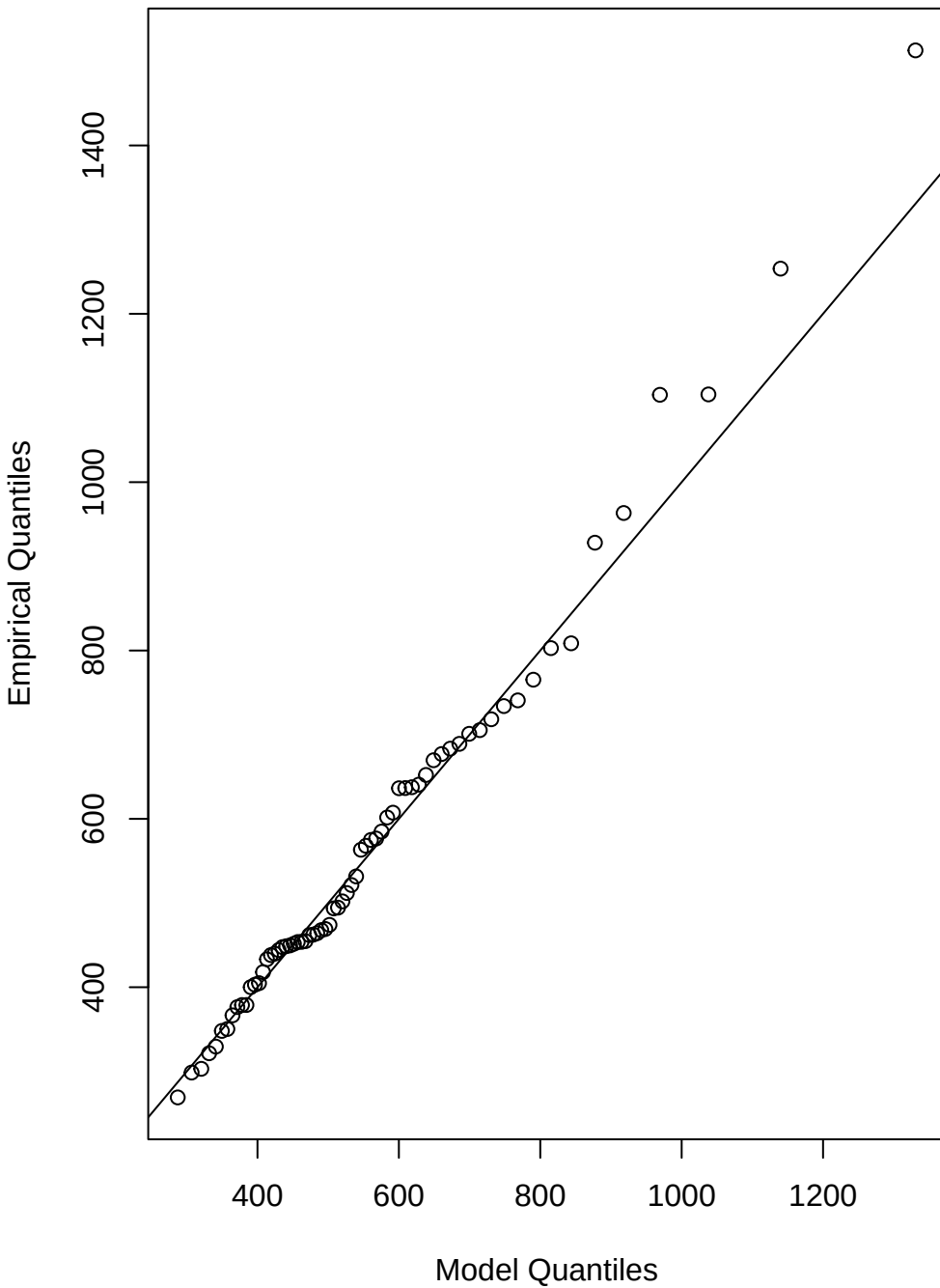


Stationary Return Level Curve: BCHLJ

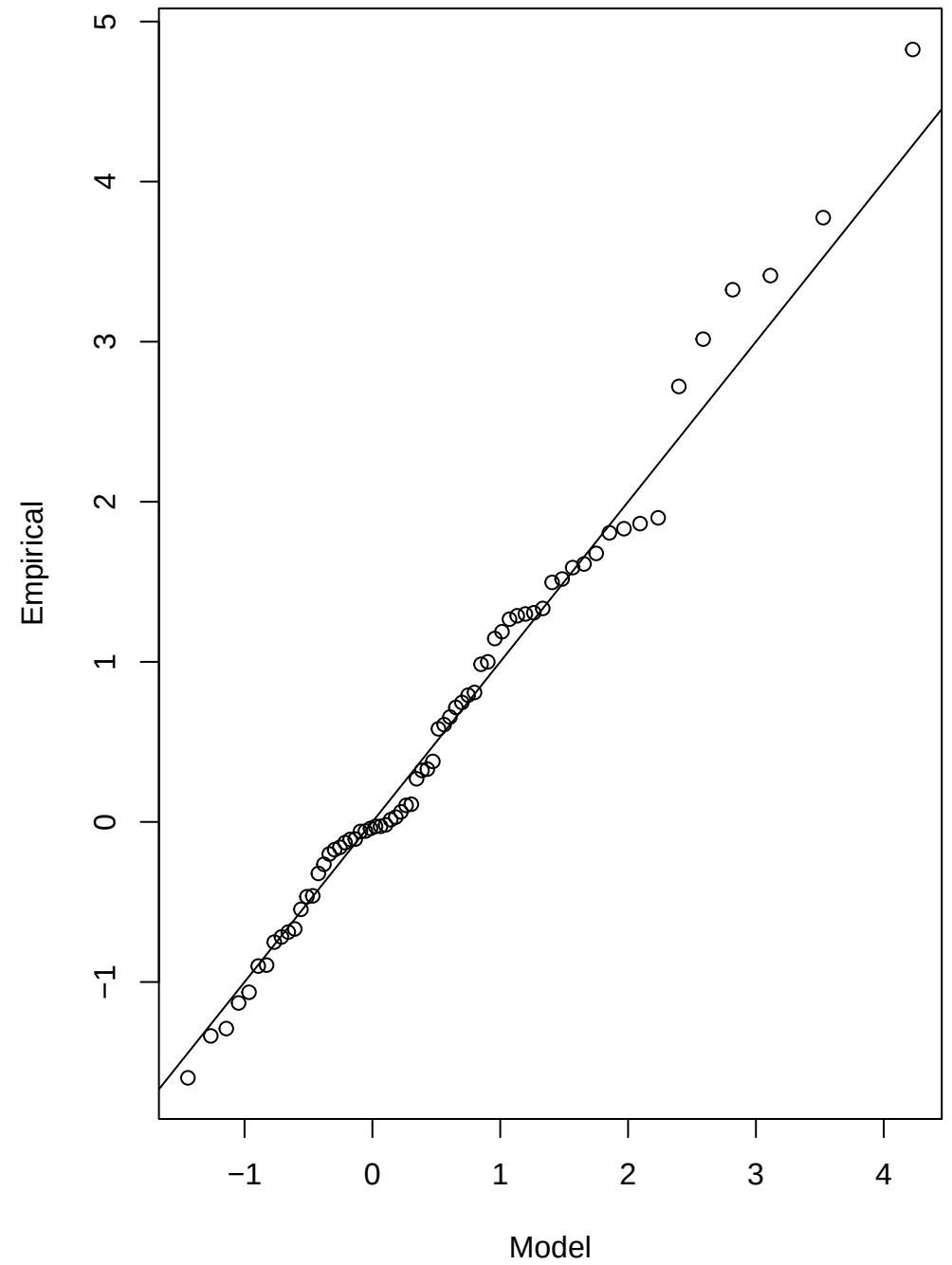


Diagnostic Fit (QQ) for Station: BCHSF

Stationary Diagnostics



Non-Stationary Diagnostics



Stationary Return Level Curve: BCHSF

